Module 1: Traditional NWP Review

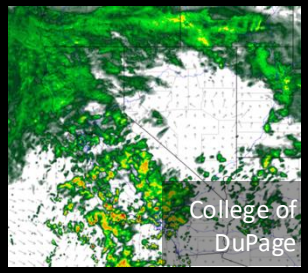
The traditional way…

Data → Analysis (assimilation)

CIRA/CSU

Run model (integration + parameterization)

Write output at pre-determined interval



The Good:
Rooted in equations of motions. Outputs are usually *physically explainable*. Physical conservation laws are required by equations.

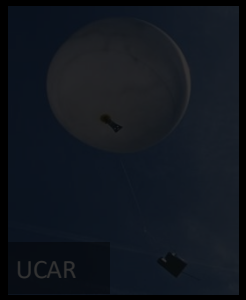
The Bad:
Speed. Depending on the computational resources, it can take hours on a very expensive compute cluster to generate a 2-week forecast.

The Ugly:
Parameterizations. Our best attempt at treating processes we cannot explicitly resolve.

ML prediction…

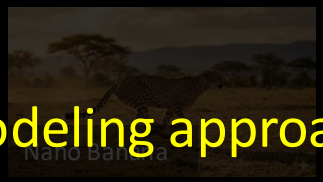
Data → Analysis (assimilation)

CIRA/CSU

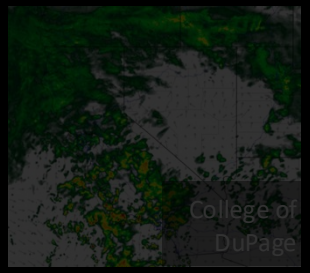

UCAR

Wikipedia

Run model (integration + parameterization)


Nano Barcelona

Write output at pre-determined interval


College of DuPage

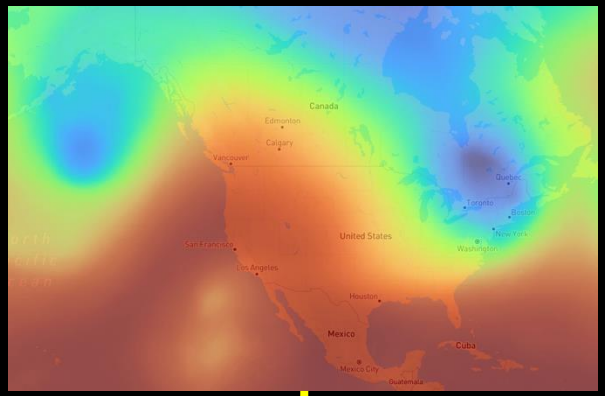
Both modeling approaches are data limited!

The Good:
Speed. Anyone with a laptop can make a 2-week forecast in minutes (after setup).*

The Bad:
Blurred predictions. MSE loss functions are penalized heavily by trying to predict a tail event, so forecasts quickly lose detail. However, significant improvement in 2024/2025.

The Ugly:
1. Explainability. Efforts are underway to improve here, but most early ML weather prediction effort have focused on performance. I.e., If a model fails, how do you explain *why* it failed?
2. No mathematical constraints on conservation laws.

*Weeks required to train as a one-time up-front expense, but then inference can be run in minutes.

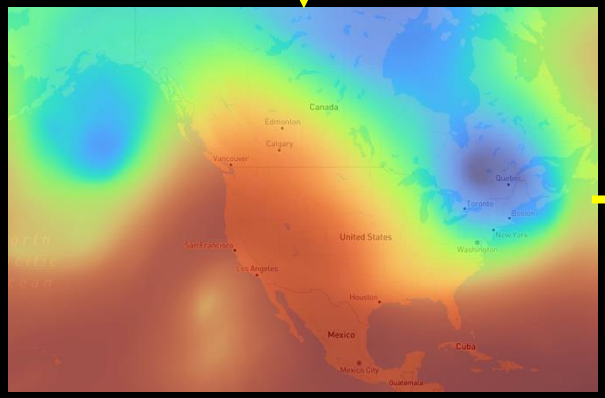


00 hr (”analysis”)

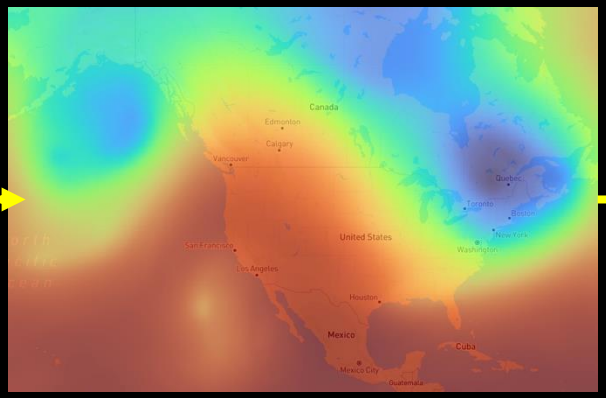
$$\frac{\partial X}{\partial t} = \cdots$$

Integrate forward in time.

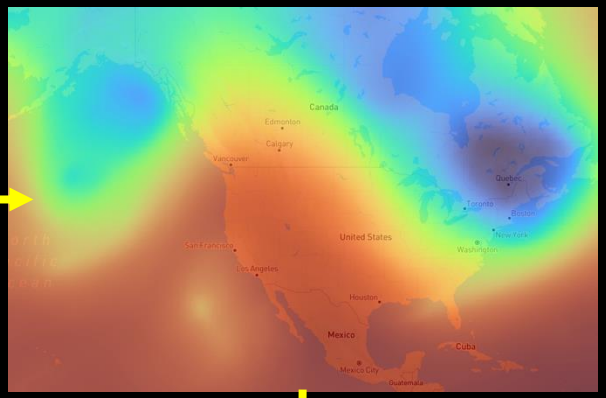
18 hr

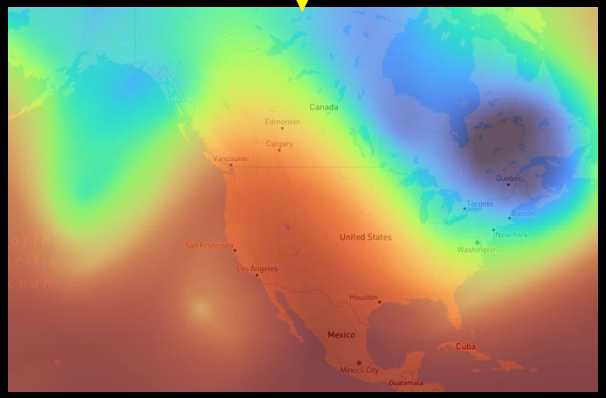


06 hr



12 hr



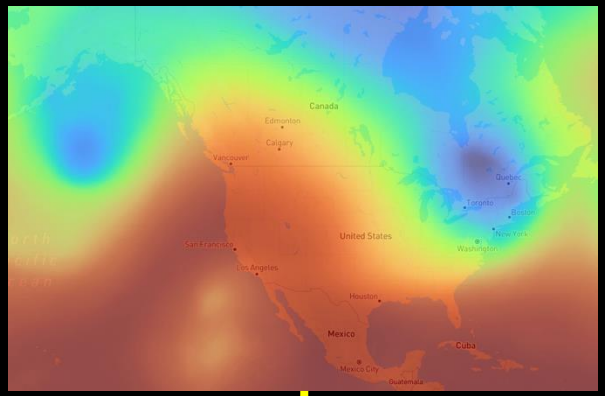


24 hr

Differ primarily in

1. Spatial discretization (How is grid handled?)
2. Temporal discretization (How does model handle time stepping?)

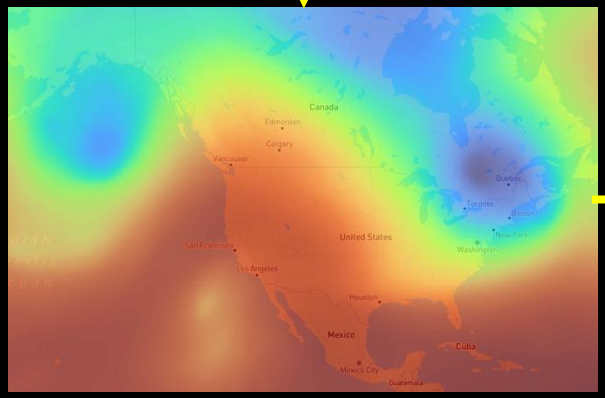
The traditional way…

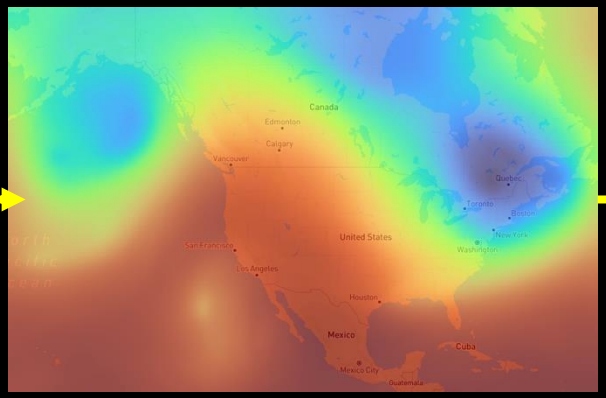


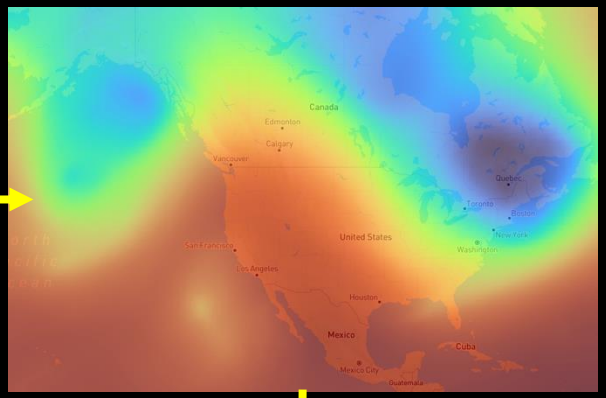
00 hr ("analysis")

Make prediction that would have minimized mean squared error in training dataset.

18 hr



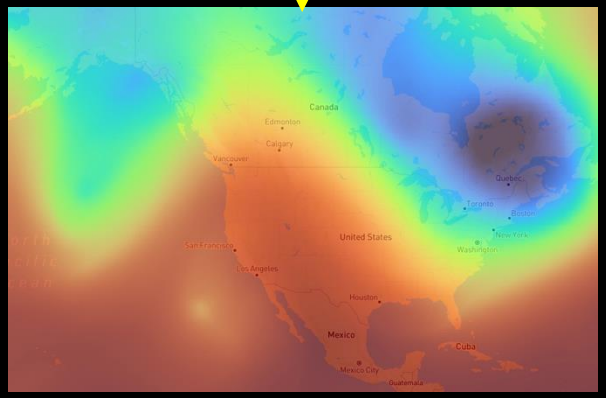




06 hr

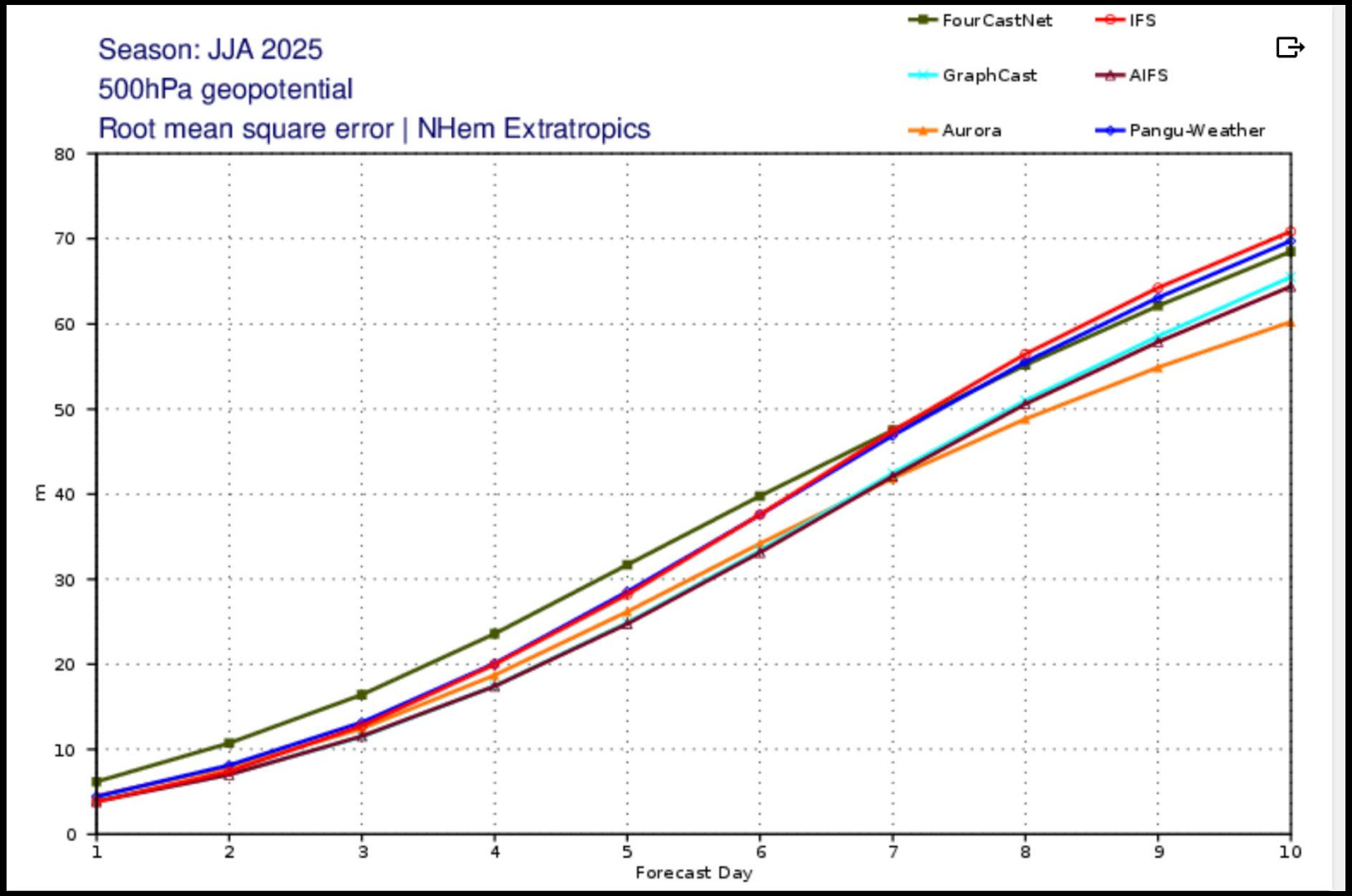
12 hr

Minimizing the loss function *and* capturing multiple scales of variability has numerous diverse methods!

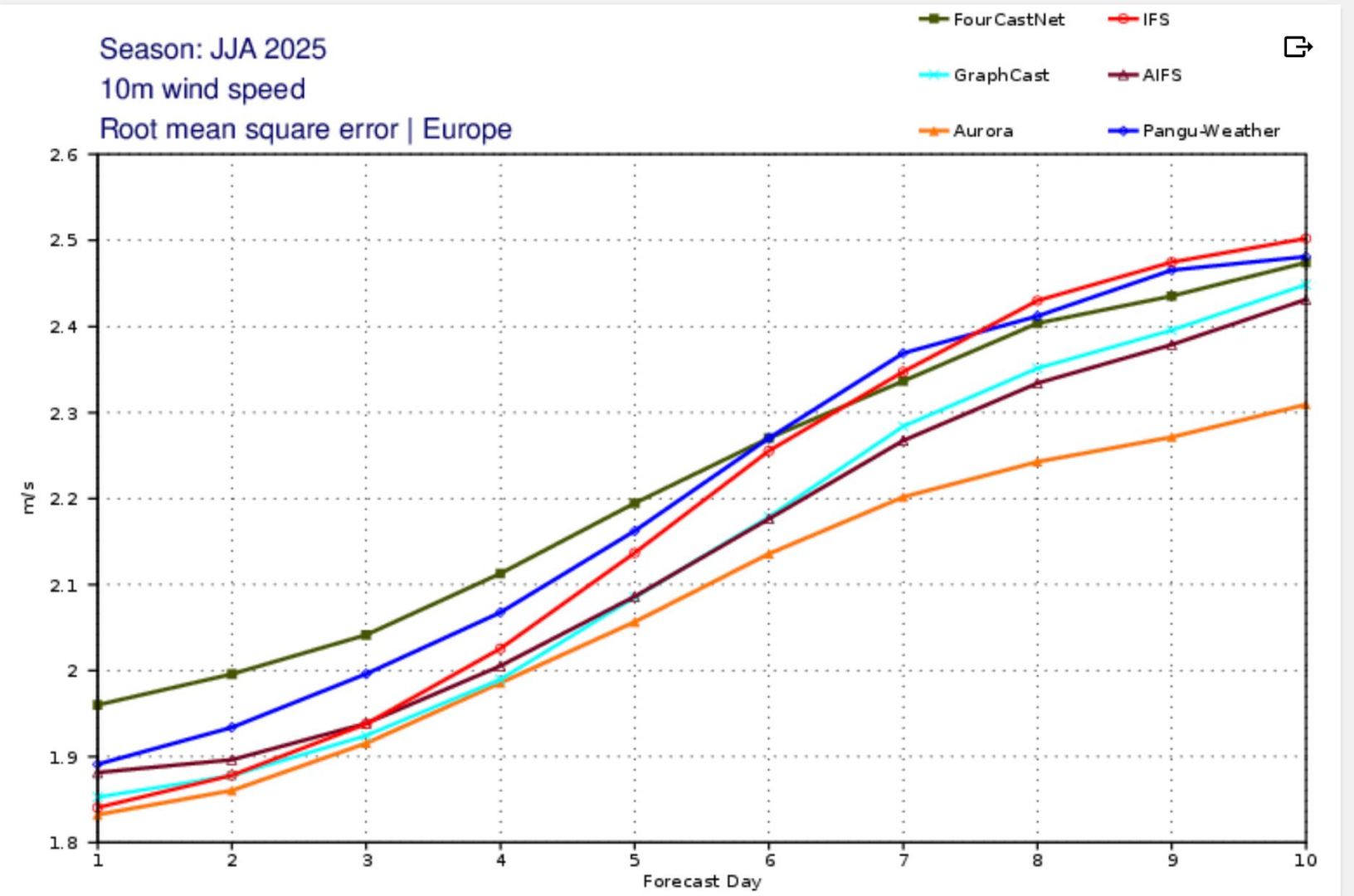


ML prediction…

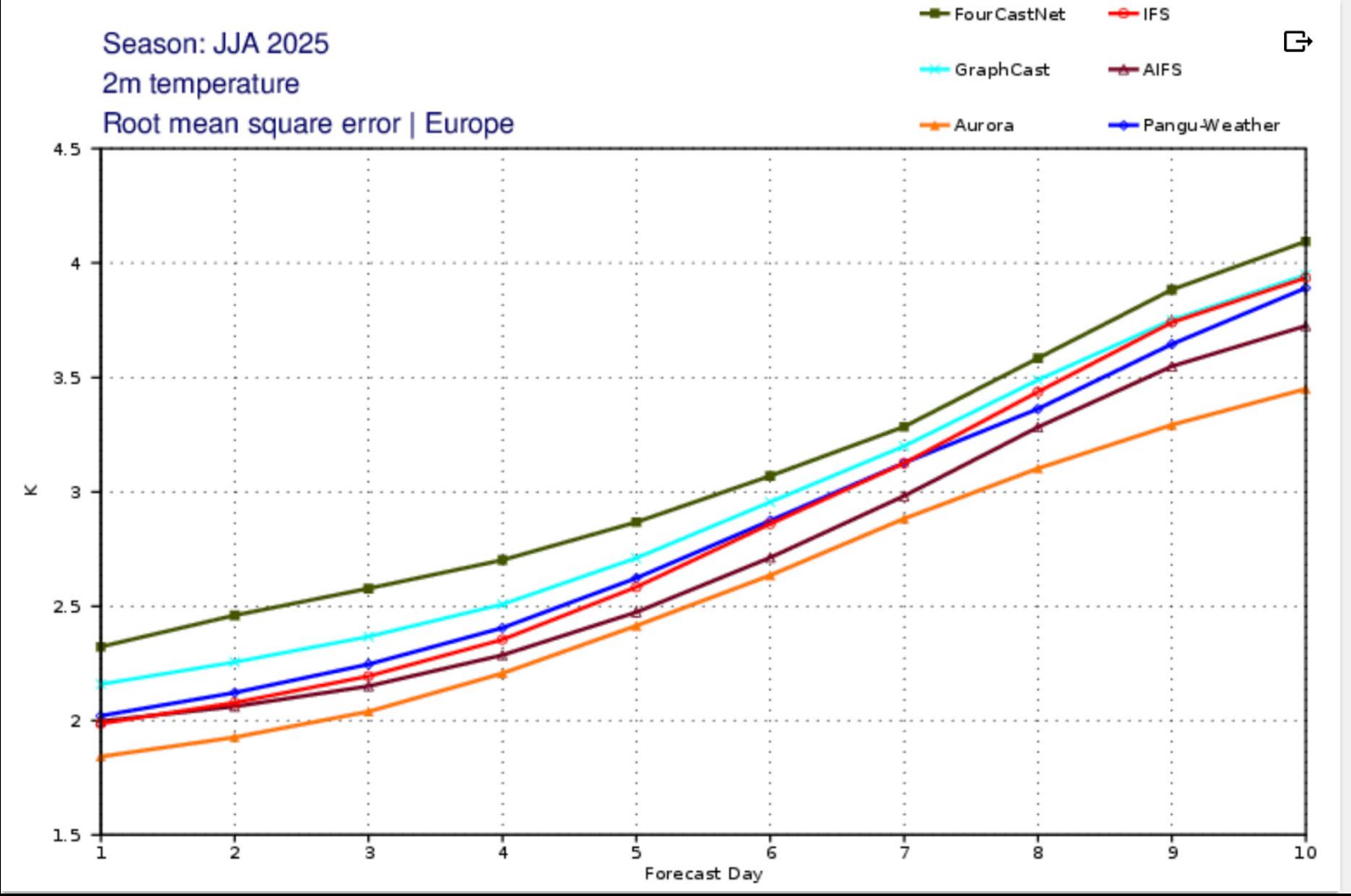
24 hr



Source: ECMWF



Source: ECMWF



Source: ECMWF

How does a gridded NWP system work?

Discretize and integrate forward governing equations.

Momentum

$$\frac{\partial u}{\partial t} = -\boldsymbol{u} \cdot \nabla u + fv - g\frac{\partial z}{\partial x} - F_x$$

$$\frac{\partial v}{\partial t} = -\boldsymbol{u} \cdot \nabla v - fu - g\frac{\partial z}{\partial y} - F_y$$

$$\frac{\partial w}{\partial t} = -\boldsymbol{u} \cdot \nabla w - \frac{1}{\rho_0}\frac{\partial p'}{\partial z} + Bg - gq$$

Mass continuity

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial \omega}{\partial p} = 0$$

Thermodynamics

$$\frac{\partial T}{\partial t} = -\boldsymbol{u} \cdot \nabla T + \omega\frac{RT}{c_p p} + \frac{\mathcal{H}}{c_p}$$

$$\frac{\partial q}{\partial t} = -\boldsymbol{u} \cdot \nabla q + E - P$$

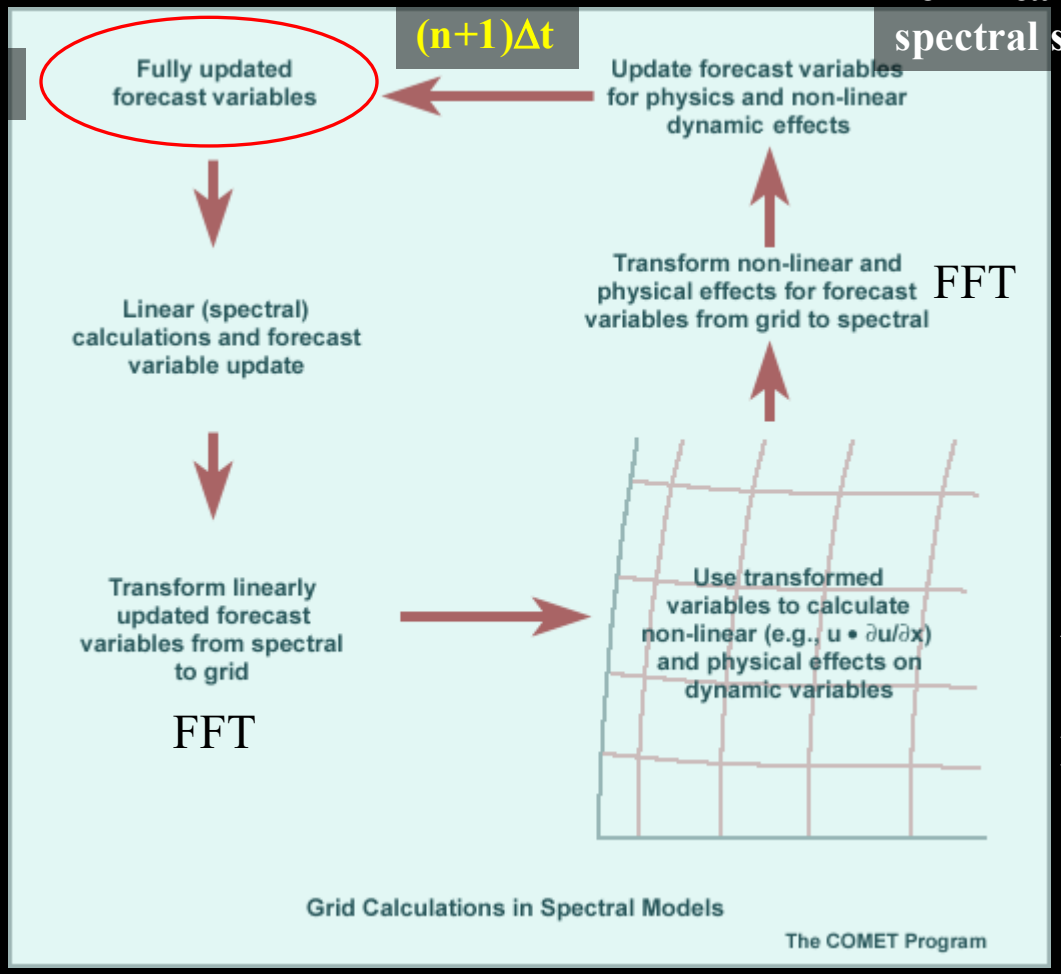
The basic challenge involves how do we step forward in time?

Flow Chart for Each Time Step

Computing final (n+1)Δt equations with linear & nonlinear terms in spectral space

return (n+1)Δt

start at nΔt

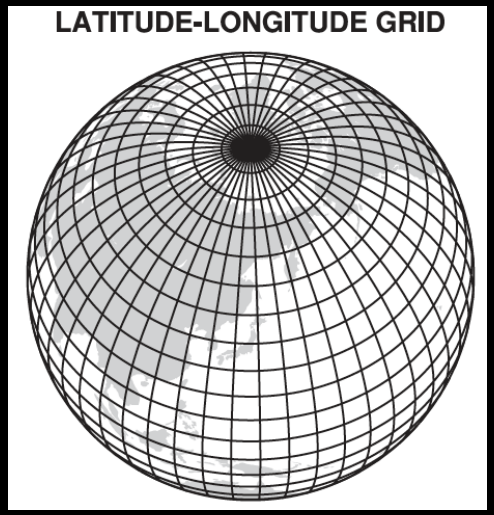


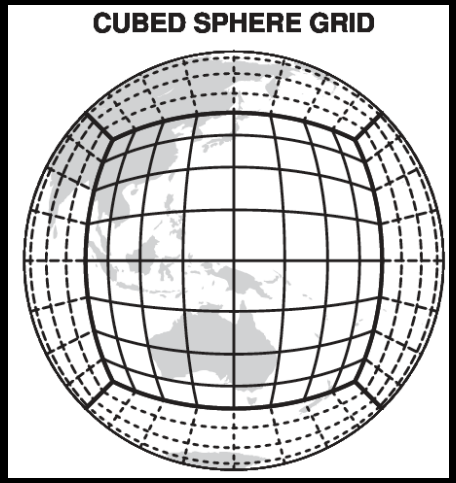
compute linear terms at (n+1)Δt spectral space

Compute nonlinear terms and physics at (n+1)Δt physical space

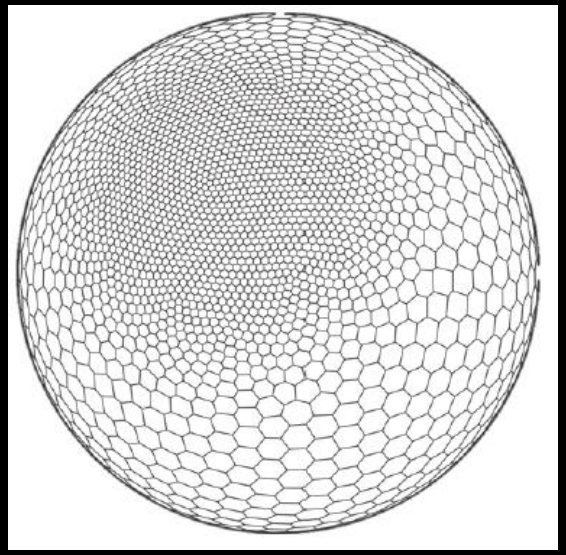
Source: COMET "Impact of Model Structure & Dynamics" http://www.meted.ucar.edu/nwp/pcu1/ic2/frameset.htm

Discretization on a grid





Hexagonal grid



Many other options available too.

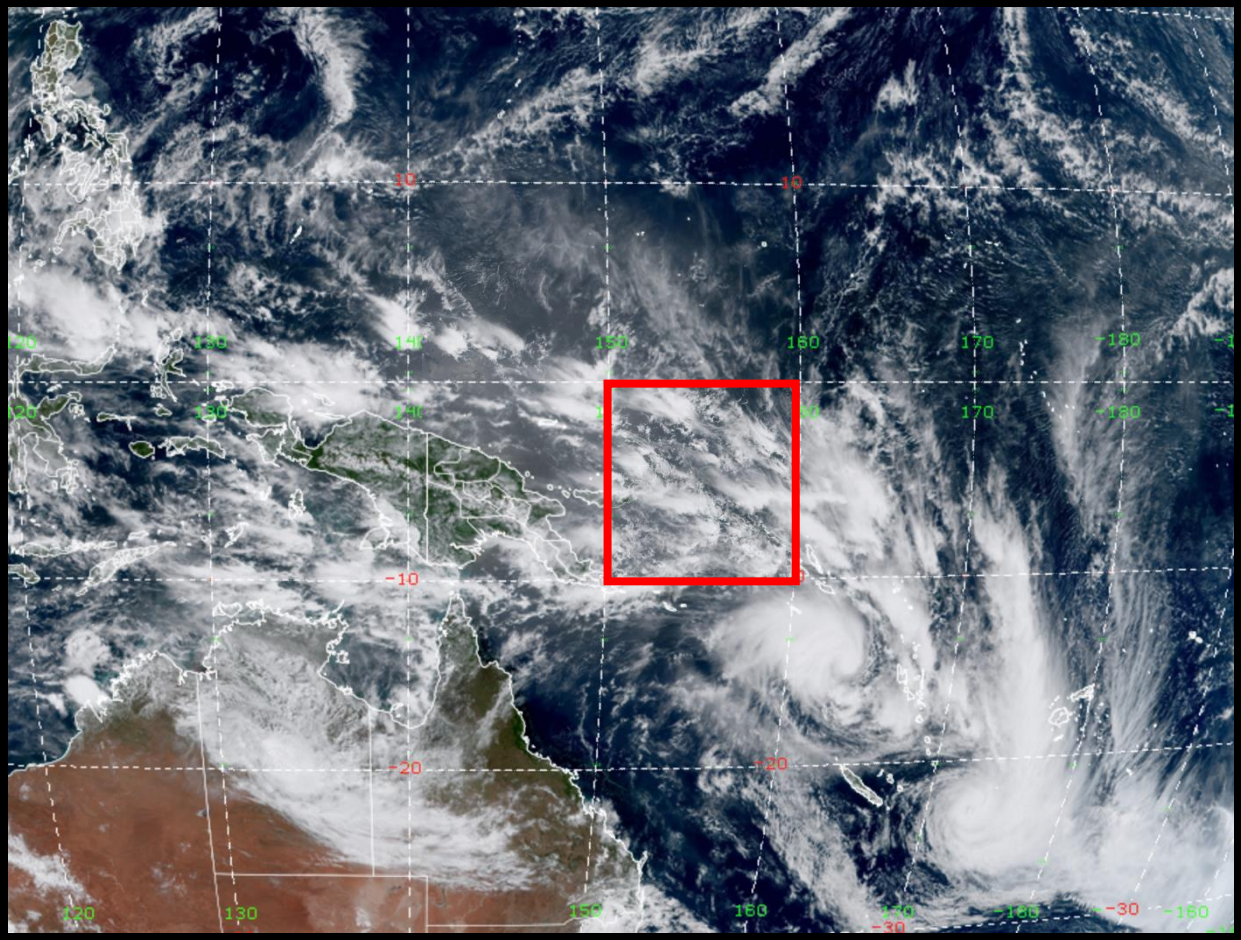


Image from Himawari-9 0200 2 March 2023 (CSU/CIRA)

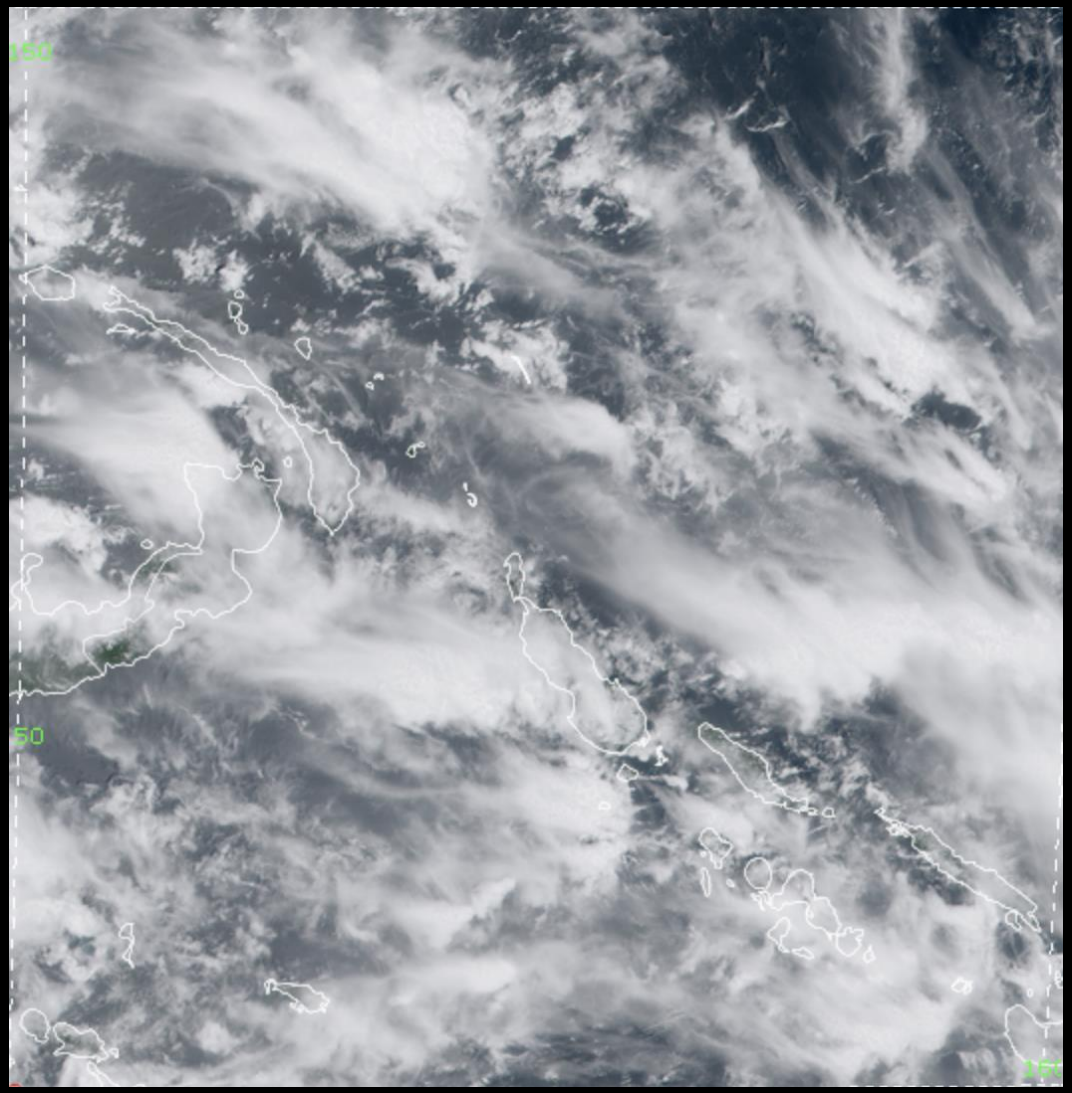
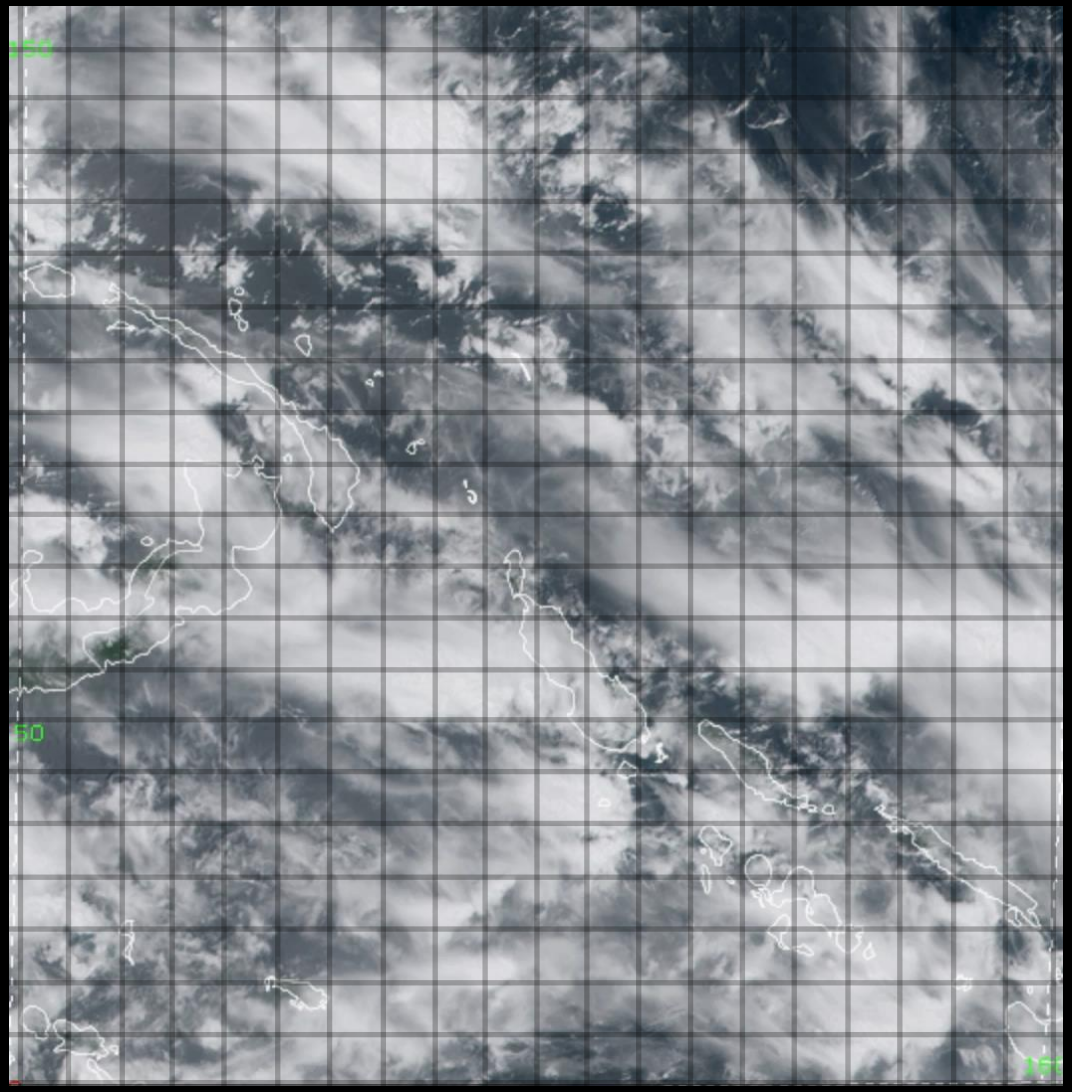
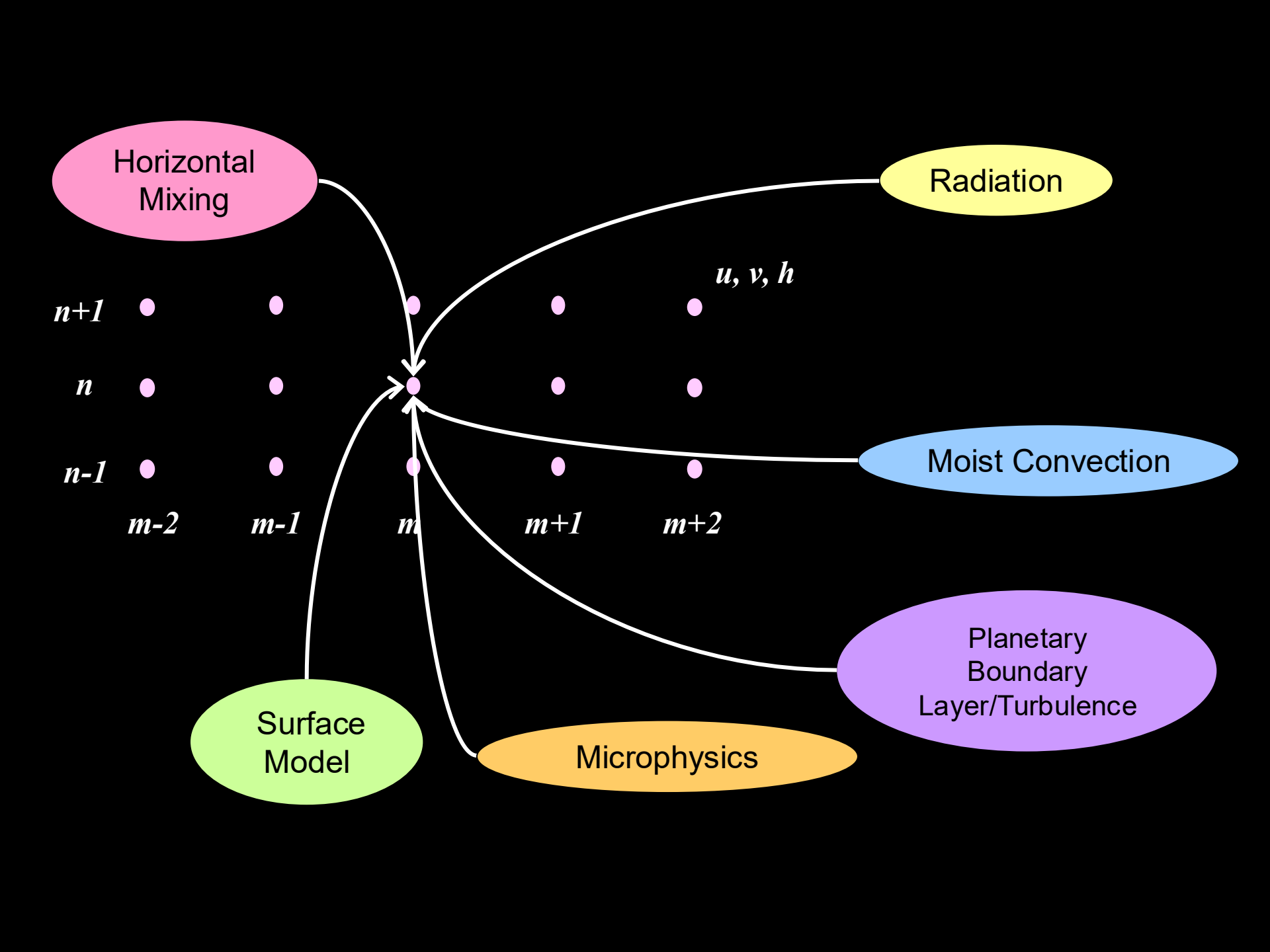


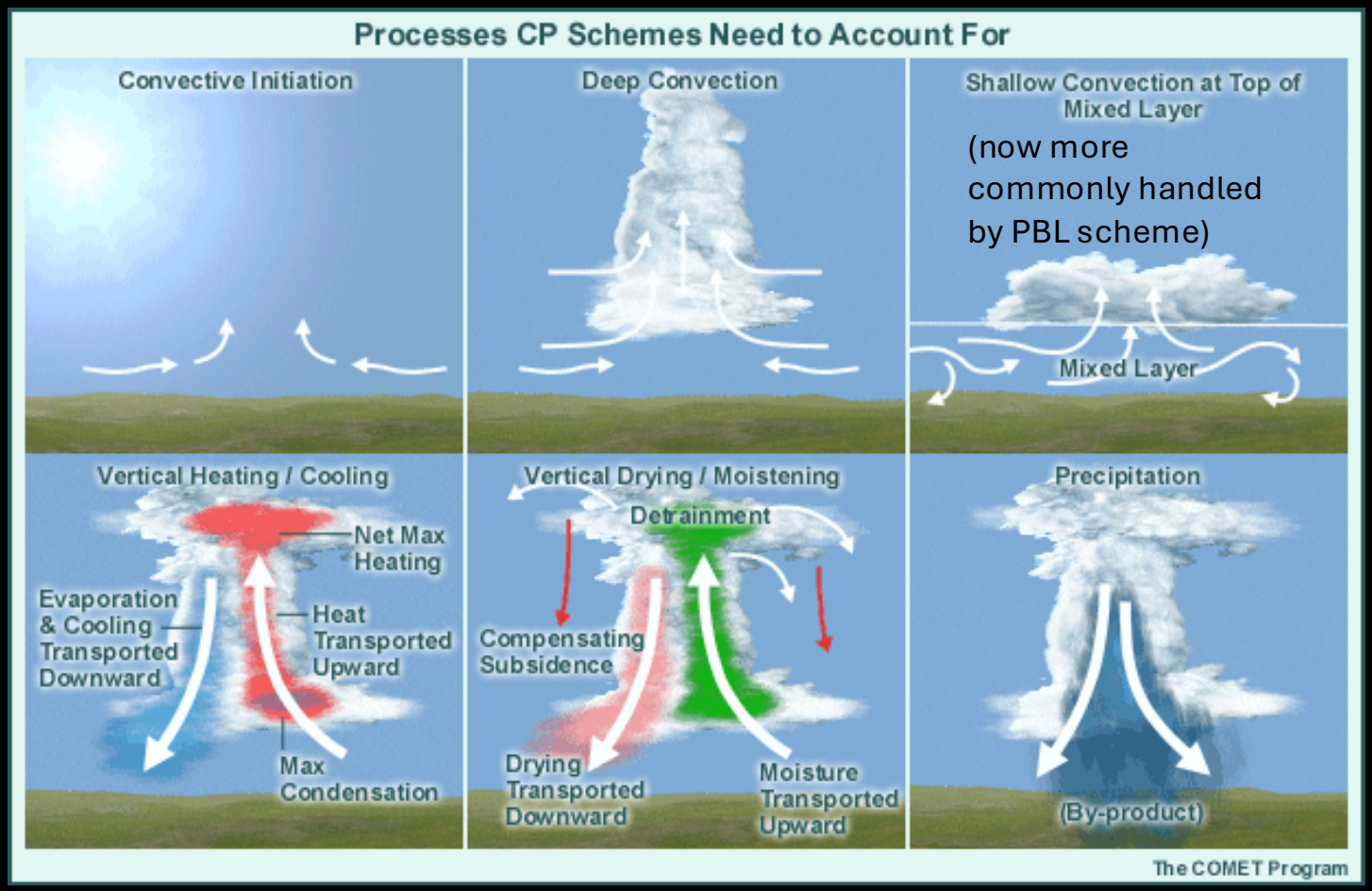
Image from
Himawari-9
0200 2
March 2023
(CSU/CIRA)



Animation from
CSU/CIRA
0200–0400 2
March 2023

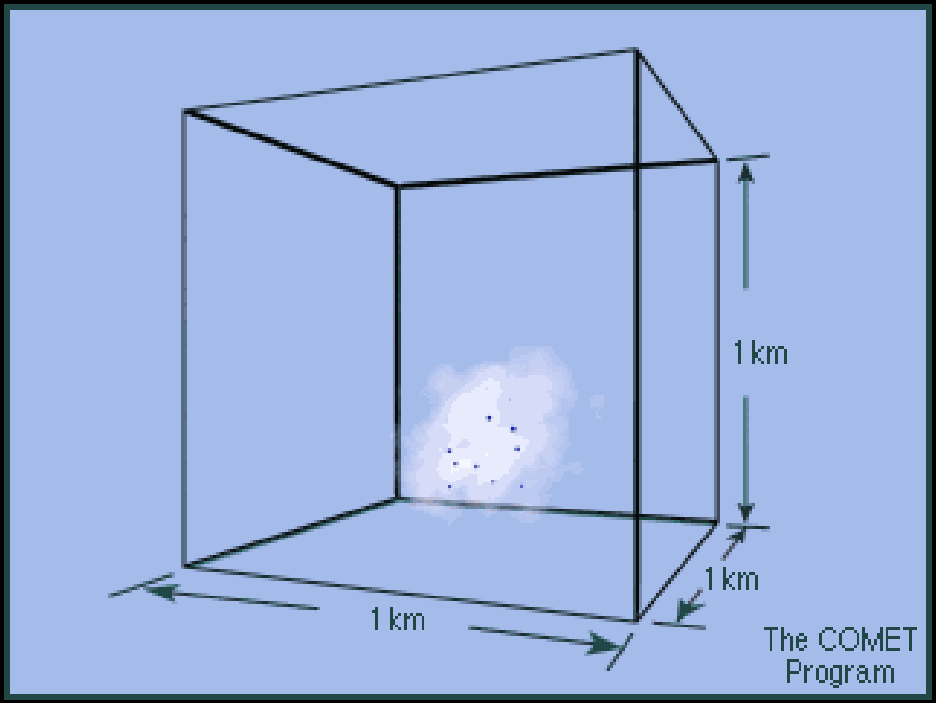

Horizontal Mixing
Radiation
u, v, h
n+1
n
n-1
m-2
m-1
m
m+1
m+2
Moist Convection
Planetary Boundary Layer/Turbulence
Surface Model
Microphysics

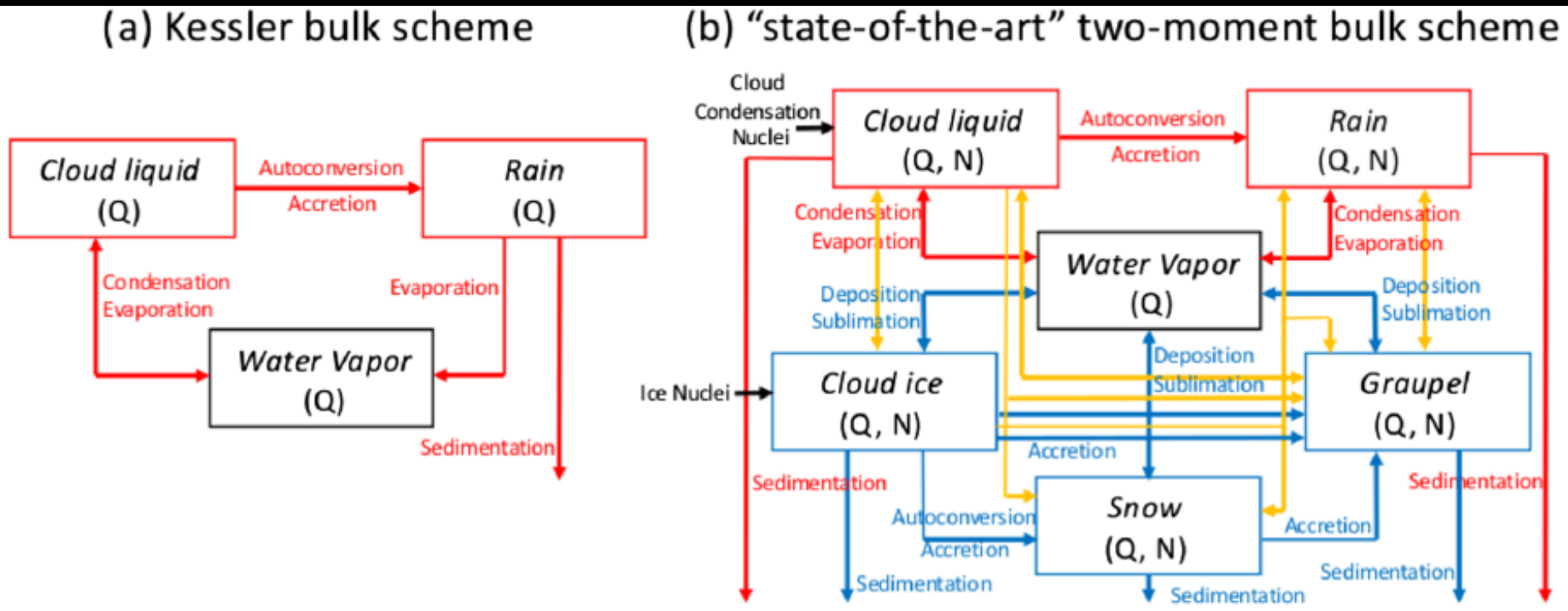
Convective Parameterization



Microphysical Parameterization

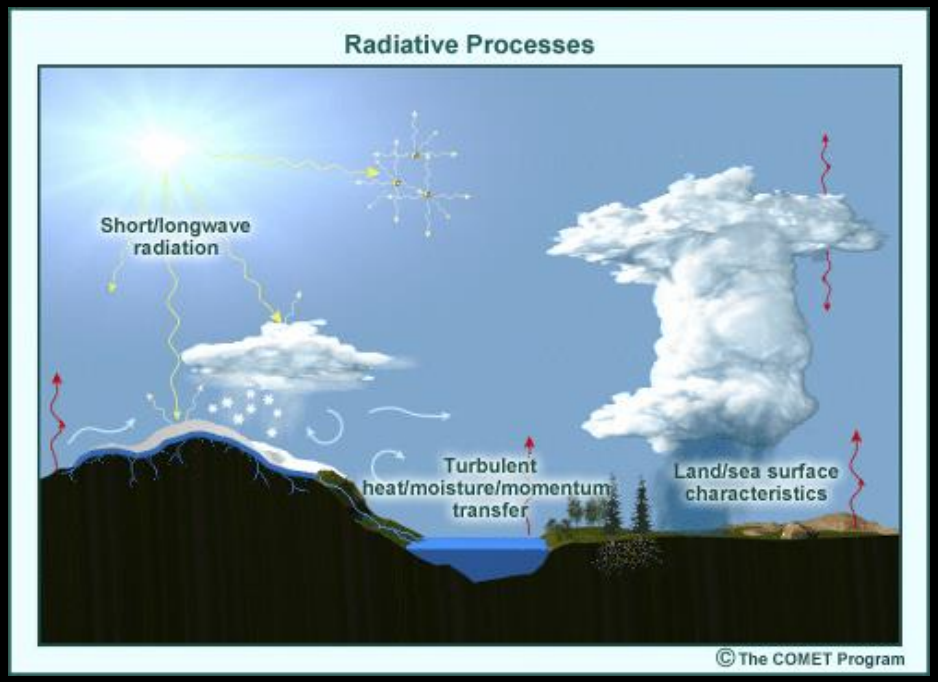
- **Microphysical processes:** Even in very high-resolution models, microphysical processes occur on a scale too small to be modeled explicitly.
- Typical m^3 of cloud has ~10,000,000 drops!
- There are important variations in both the horizontal and vertical.
- In the animation, condensation and droplet growth are occurring inside a 1-km model grid box

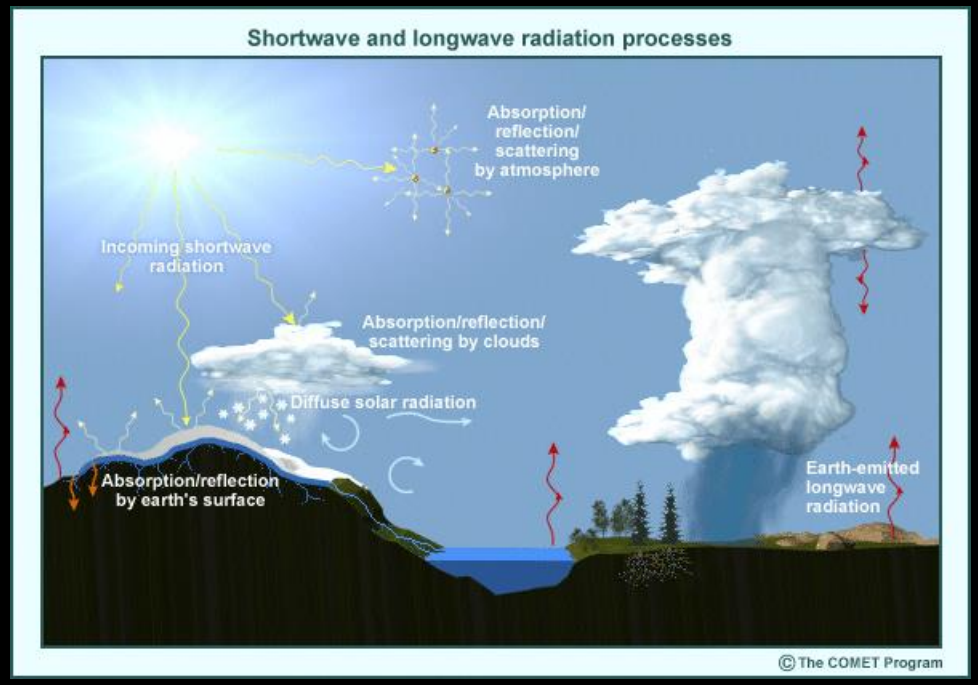




Morrison et al. (2020)

The physics of radiation are unresolvable at grid scales (photons!).



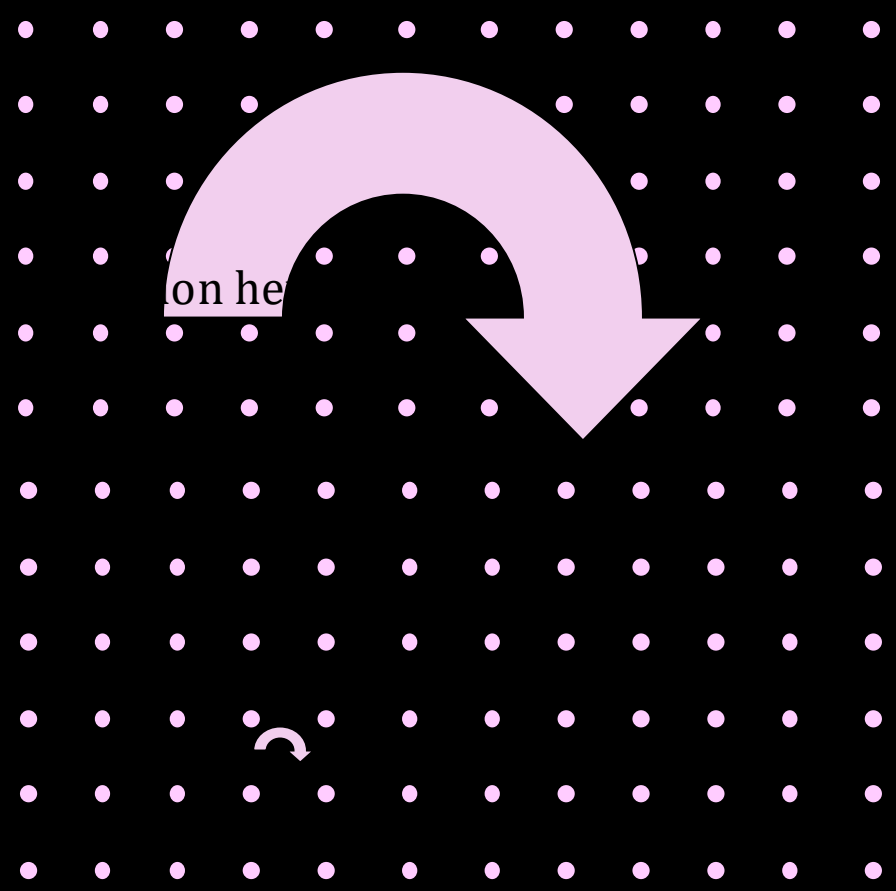


Radiative processes must include:

- Shortwave (solar/VIS) and longwave (terrestrial/IR) radiation processes that occur on very small time/space scales and are strongly affected by the local composition of the atmosphere:
  - Absorption by various gases, clouds, and aerosols.
  - Reflection and scattering by all gas molecules, clouds, and aerosols.
- Land and sea surface characteristics which affect albedo, radiation absorbed or emitted by the surface, evaporation, sensible heating
- Exchange of heat, moisture, and momentum between the surface and atmosphere

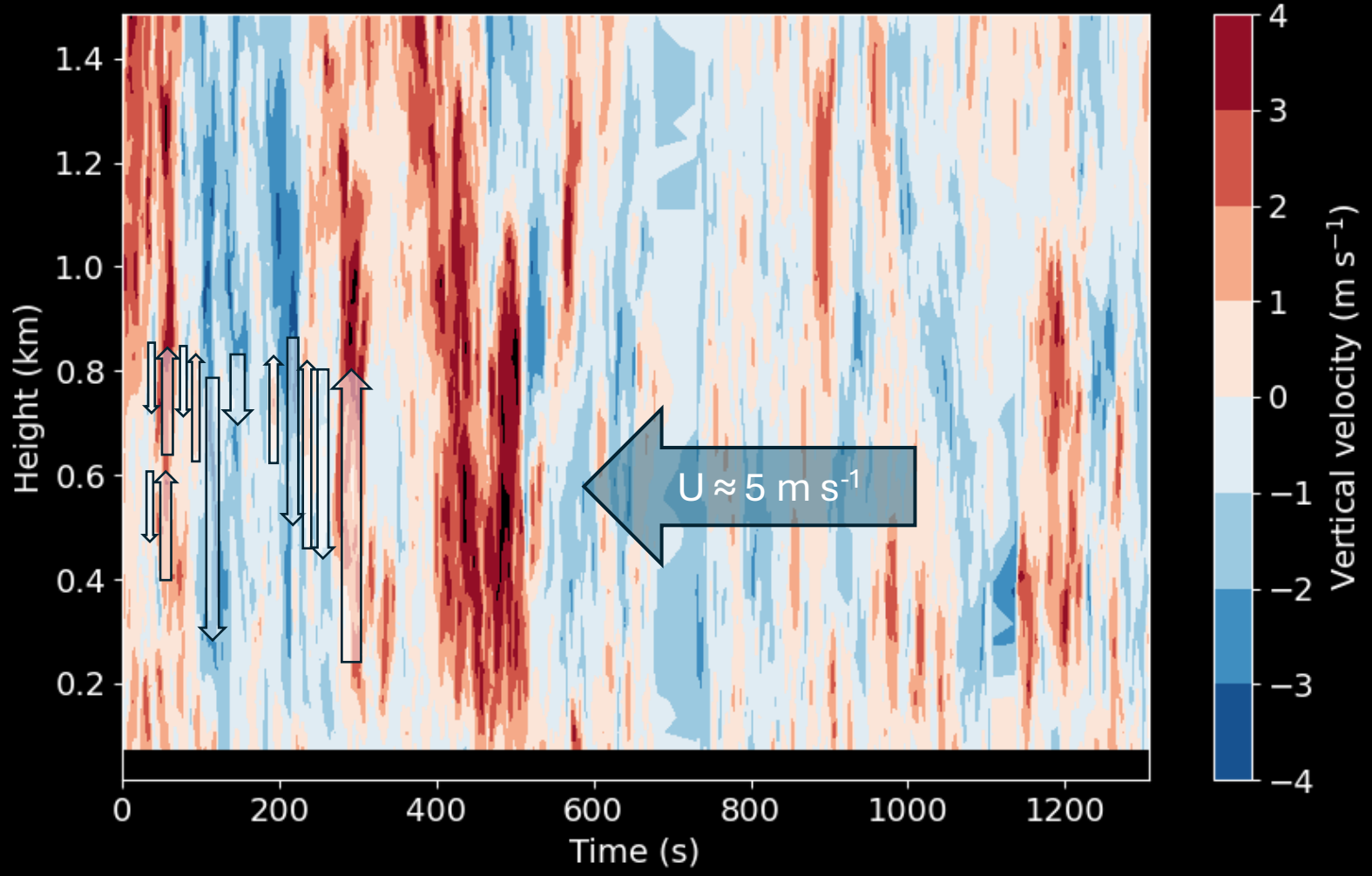
Boundary Layer Parameterization

- Mixing caused by eddies of scale ~6Δx and larger is <u>resolved</u>.
- Mixing by eddies too small to resolve is "subgrid" mixing and its <u>effects</u> must be parameterized.
- We will call subgrid scale eddies <u>turbulence.</u>



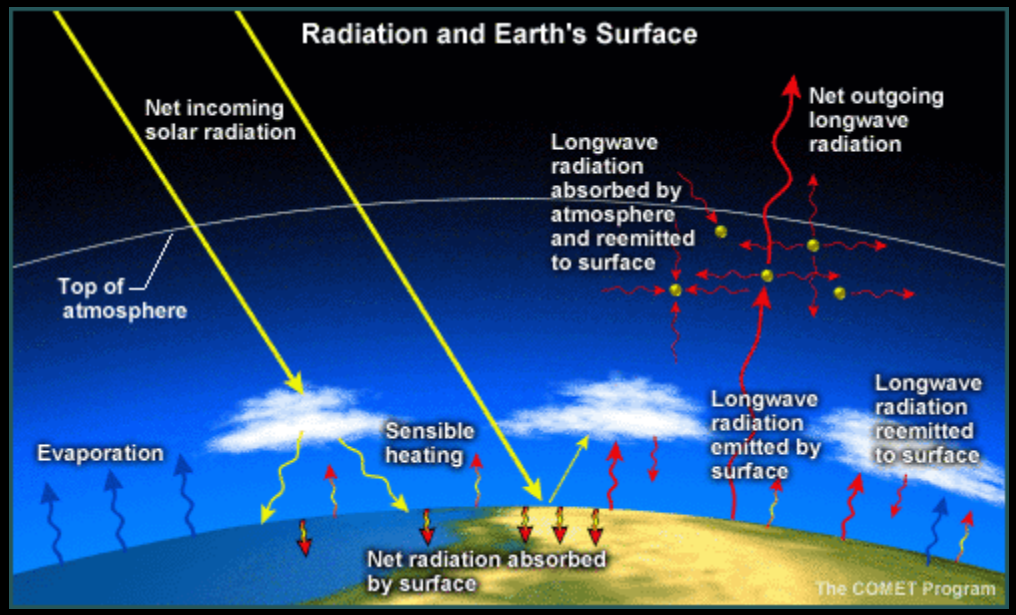
$$u = \bar{u} + u'$$

This is what the energy containing portion of real turbulence can look like in the boundary layer from perspective of upward viewing Doppler lidar.

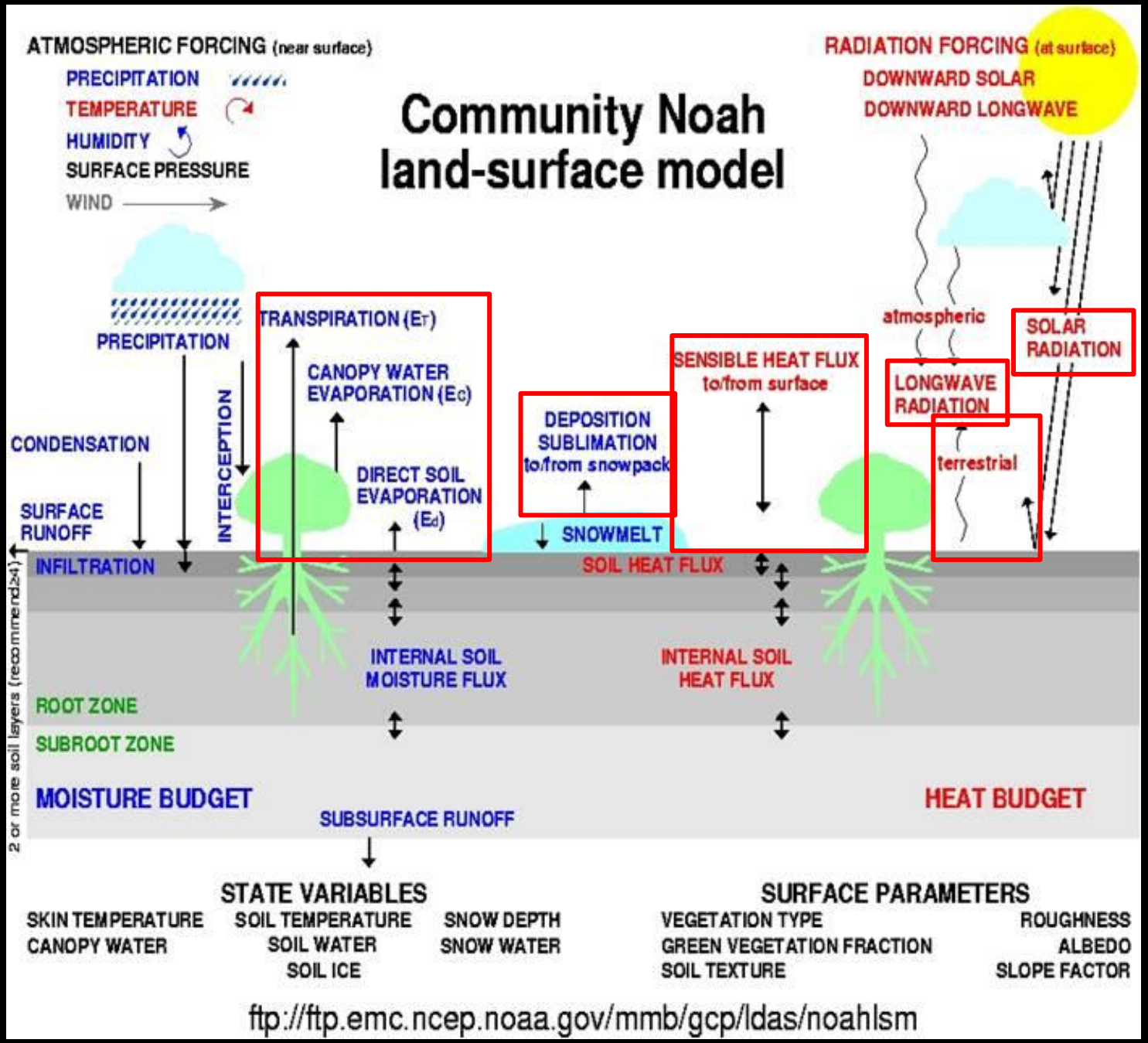


For flow of 5 m s^{-1}, energy containing eddies are O(100m) across. Not resolved in models *or* observations.

Land Surface Parameterization



- Surface processes are important to model properly, because they determine the net surface energy balance, which determines surface temperature and PBL characteristics.



ftp://ftp.emc.ncep.noaa.gov/mmb/gcp/ldas/noahlsm

Gravity Wave Drag Parameterization

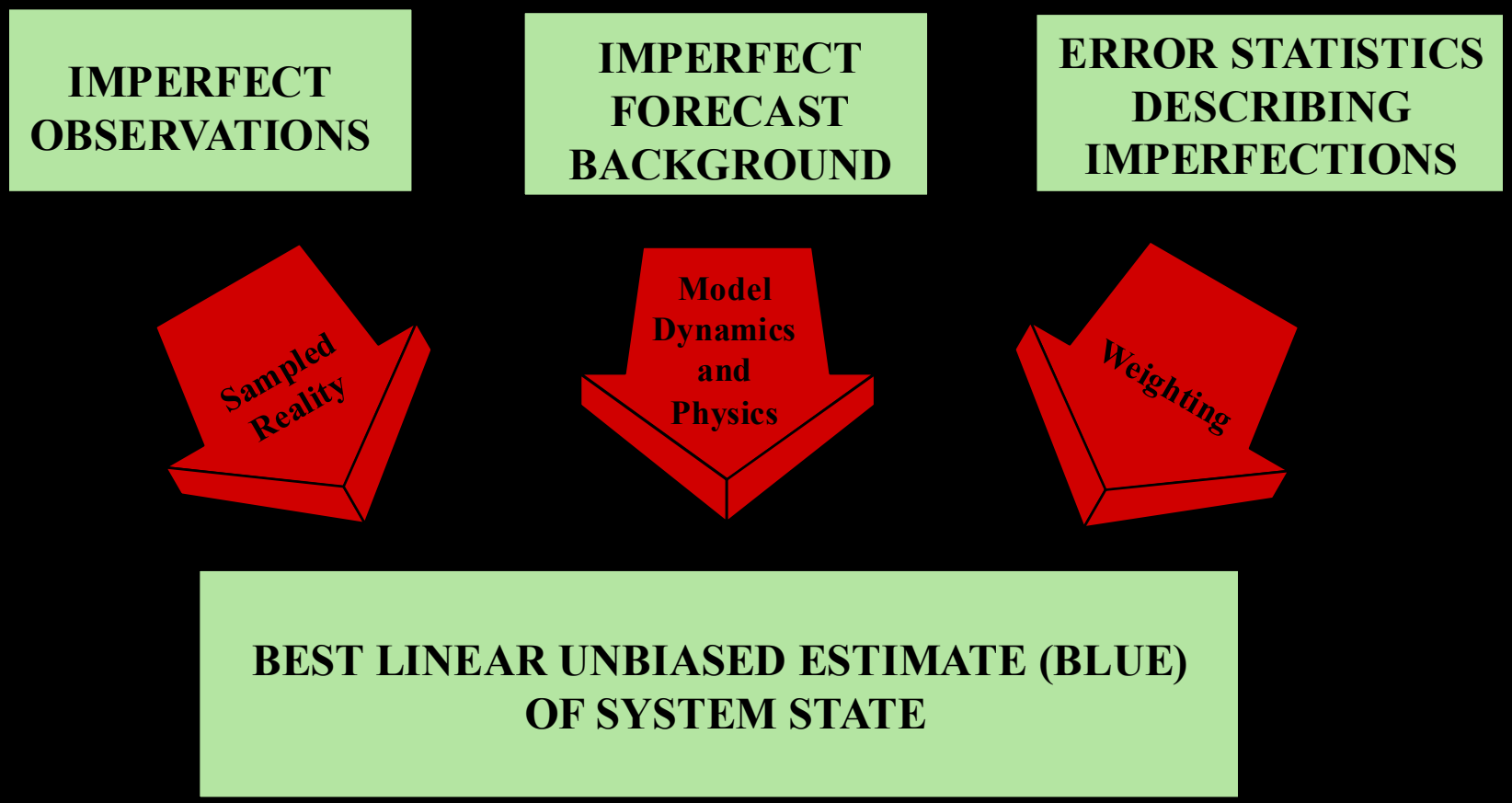
- Numerical models typically require a parameterization for the effects of sub-grid scale gravity waves on the grid-scale flow

- Sub-grid-scale orographic or non-orographic gravity waves can propagate vertically and deposit their momentum when then they break. This leads to acceleration or drag on the grid-scale flow

- Gravity wave drag parameterization is designed to approximate the effect on the resolved flow of the gravity wave drag that would be generated by unresolved subgrid-scale topography or non-orographic sources in an atmospheric model

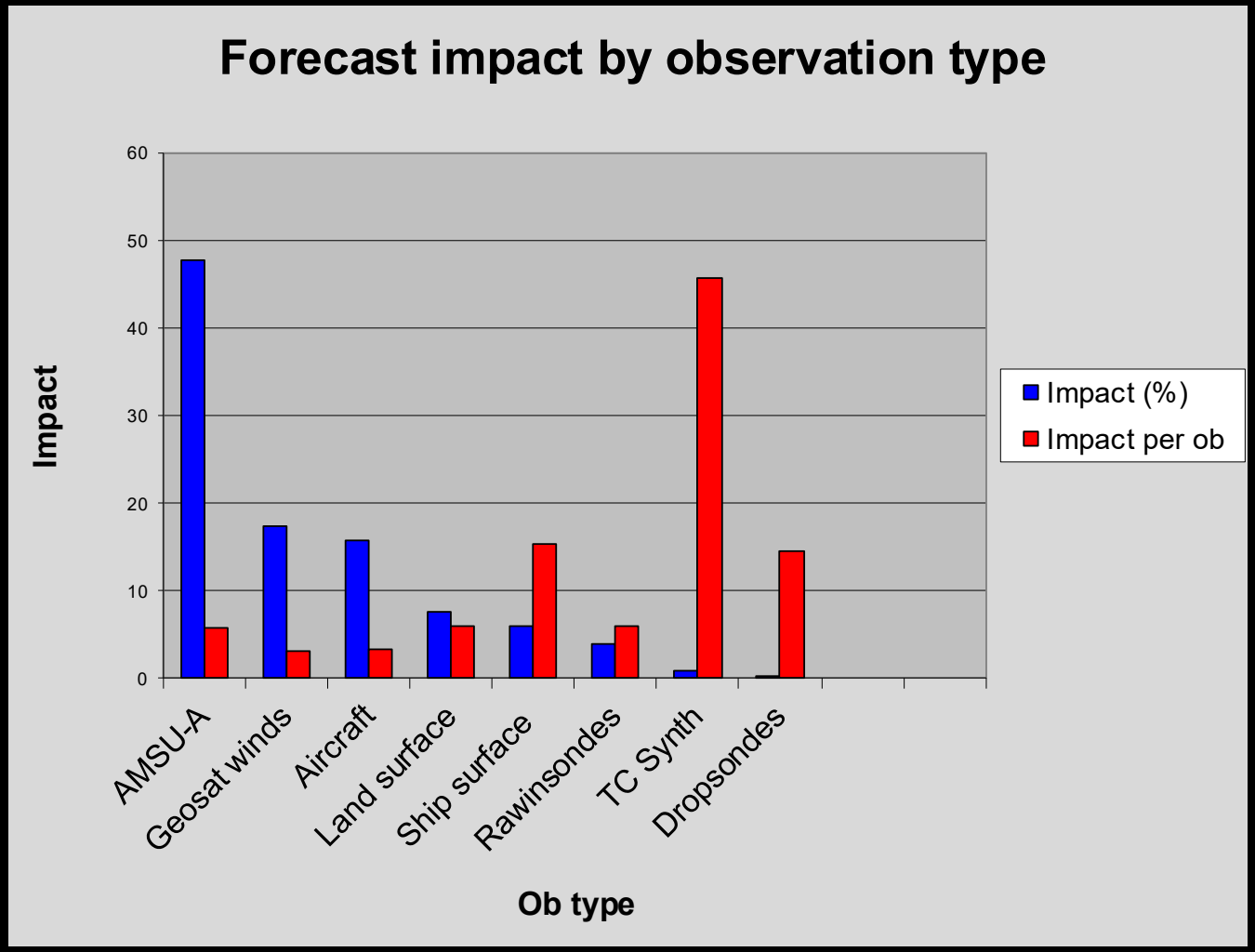
Model	Observations
Gridded	Irregular
Resolves only grid scale features; subgrid scales are parameterized	Observation scales vary; may resolve subgrid processes
Balance constraints, maybe including hydrostatic balance	Unconstrained and independent
Consistent with previous model state	No memory of previous observations
Model realistic, but diverges from truth without observations	Real-world measurements with observation and representativeness errors

Three Sources of Information



Quality Control

Does it work? Measure to see.

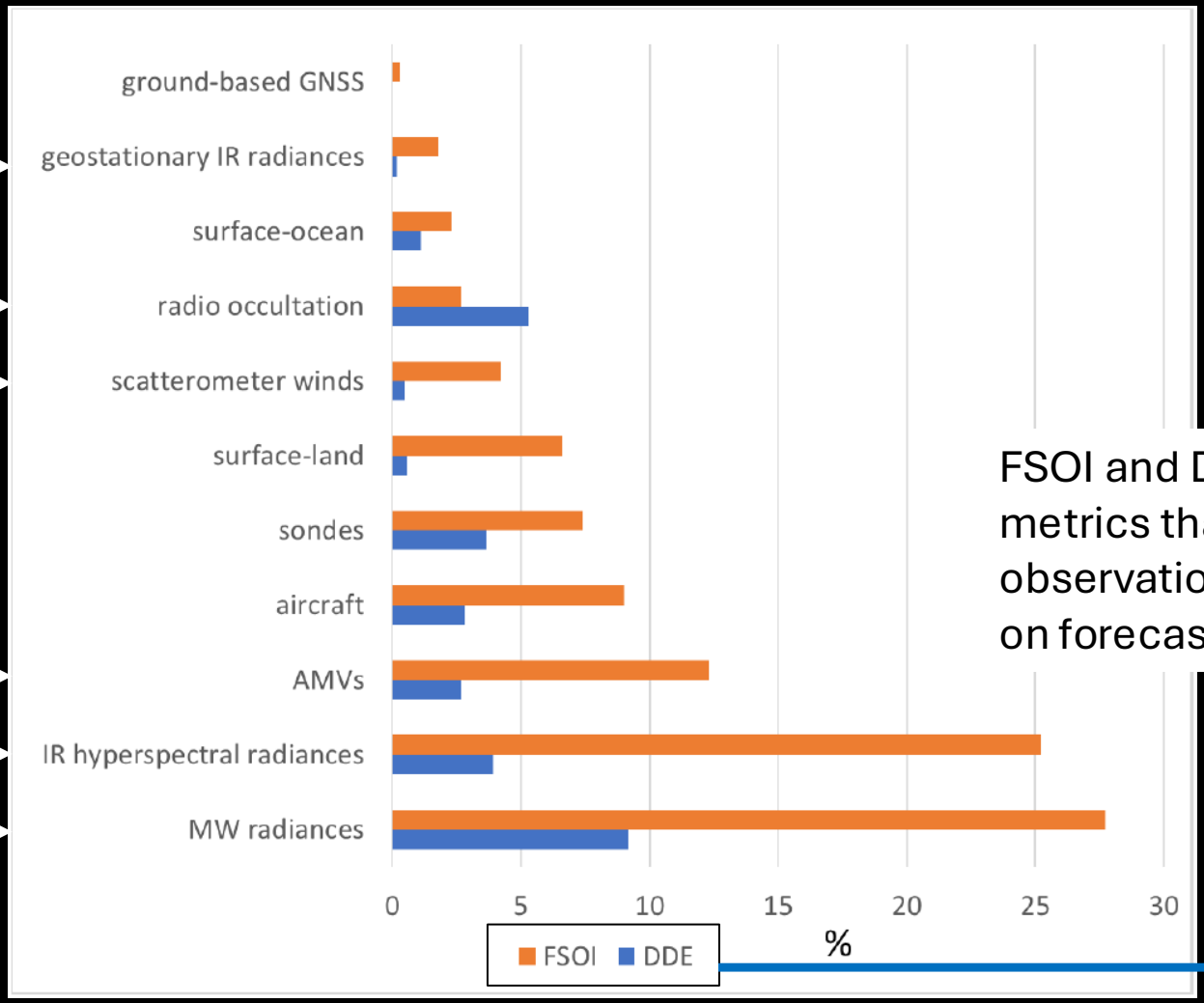


Verification by self-analysis (42 vs 48 hr. forecasts), see Langland and Baker, 2004 (*Tellus*).

Quality Control

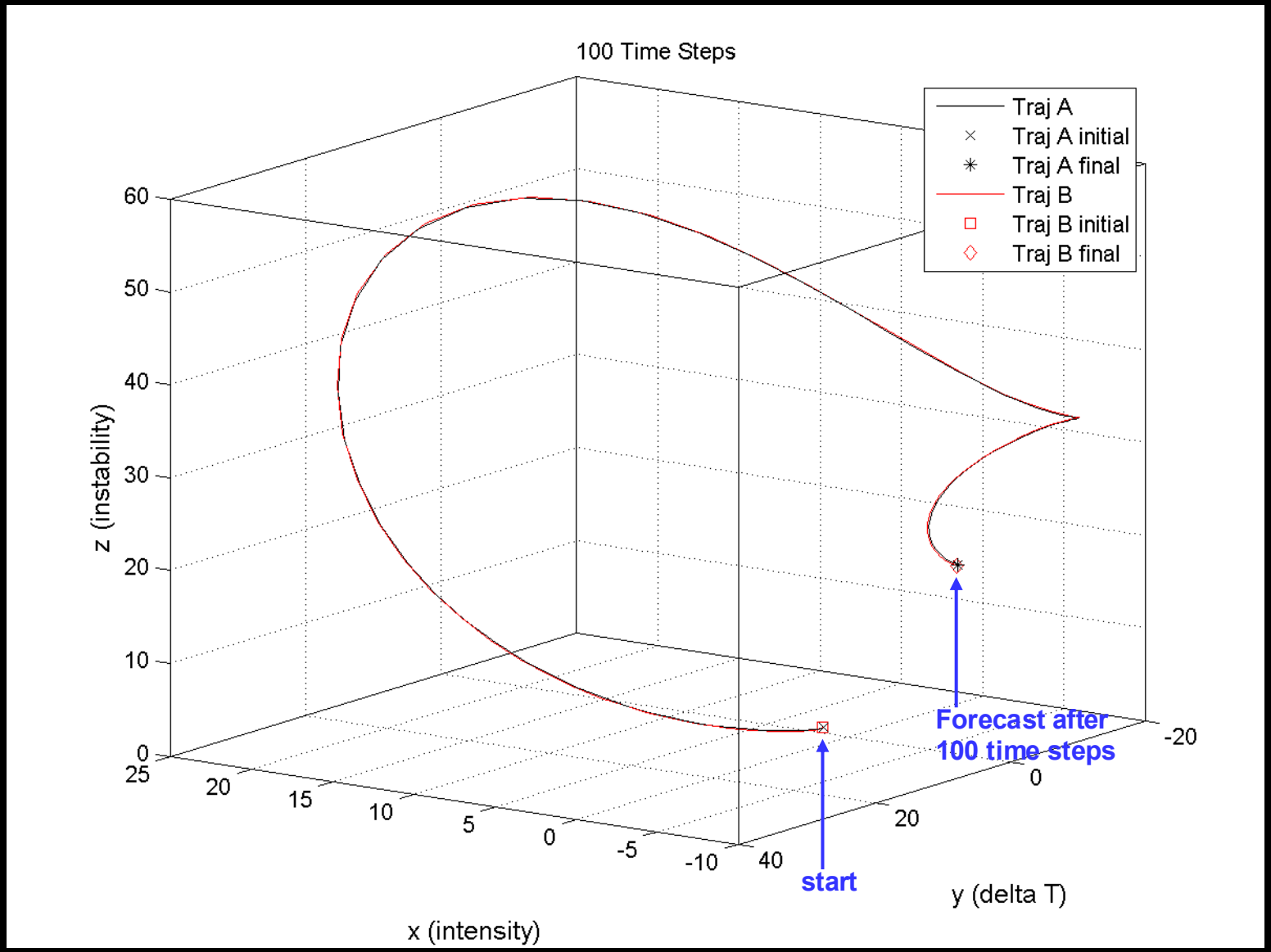
Impact on UKMO UM

Arrow denotes ob from satellite

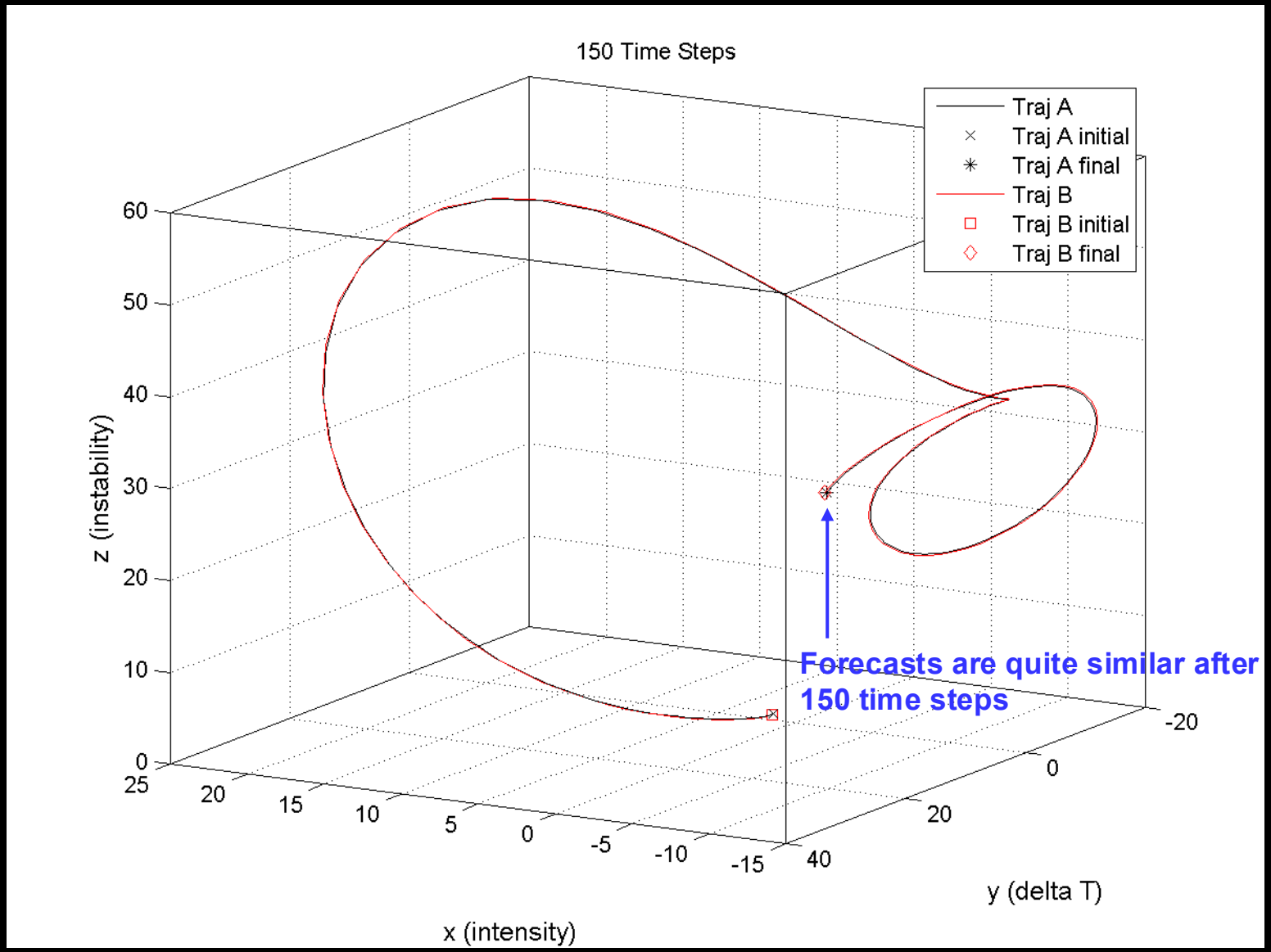


FSOI and DDE are two metrics that quantify observational impact on forecast error.

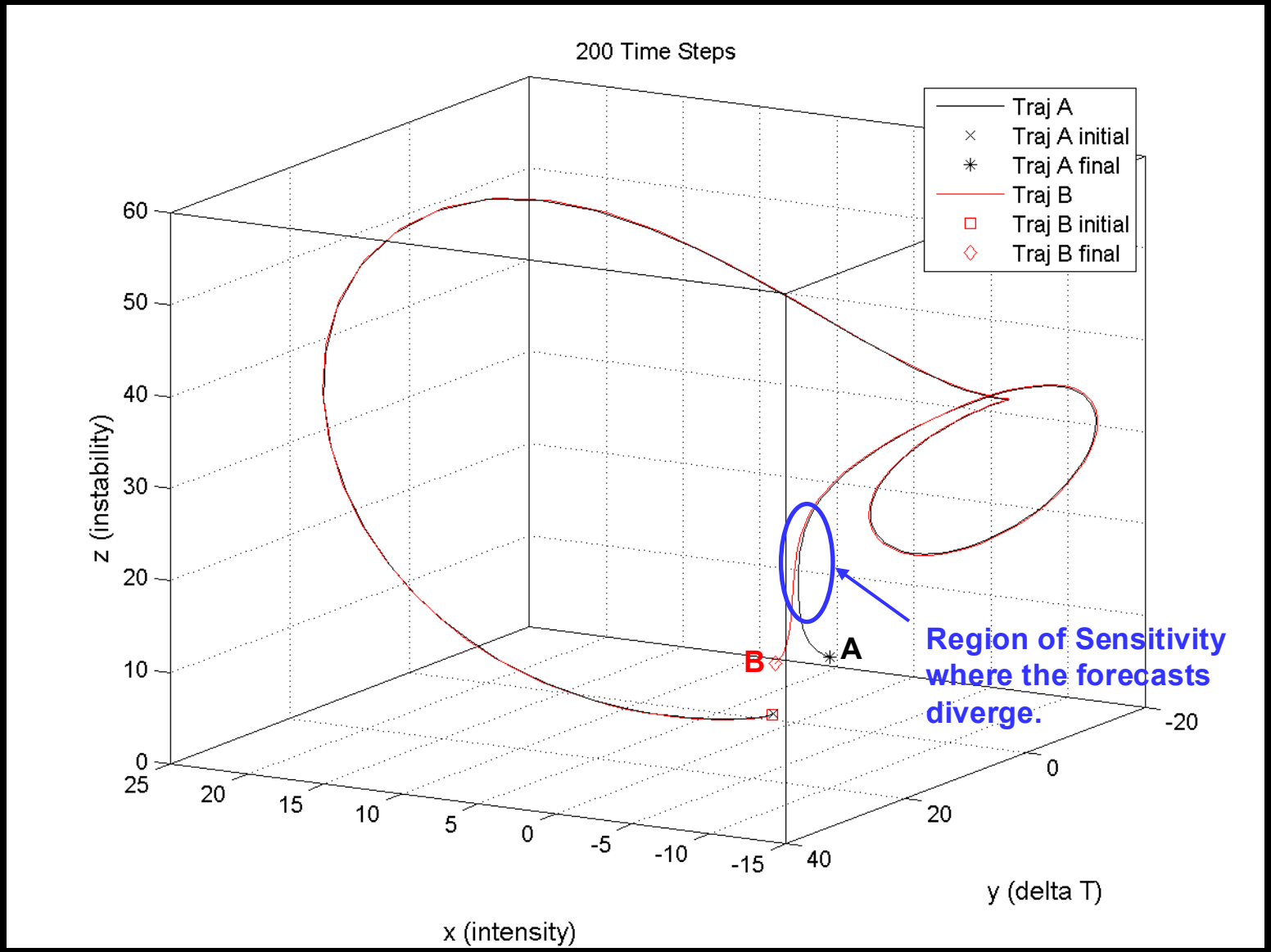
Eyre (2021), Metrics for assessing the impact of observations in NWP: a theoretical study. Part I: optimal systems



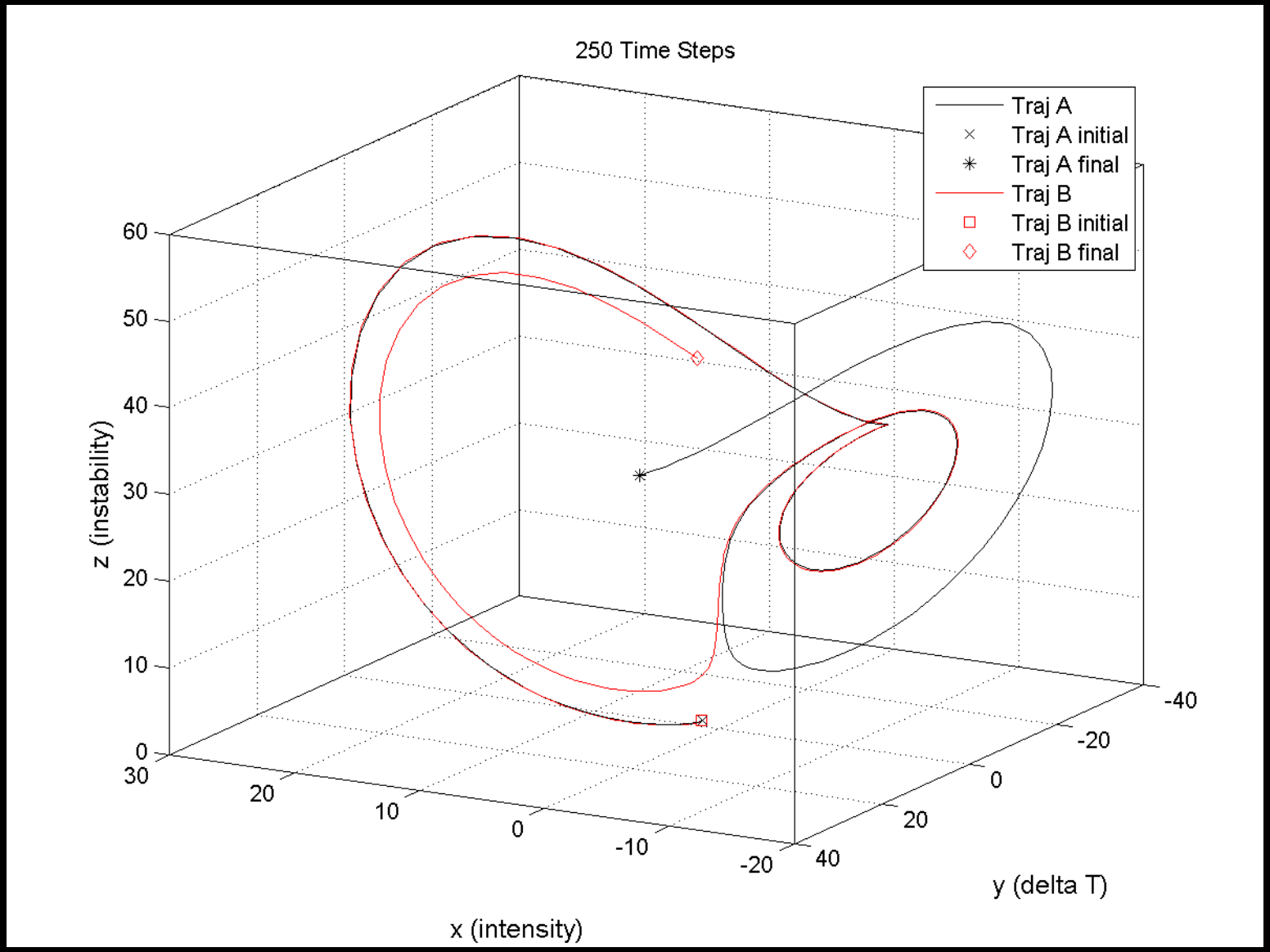
Trajectory B starts with slightly different initial conditions than **Trajectory A**

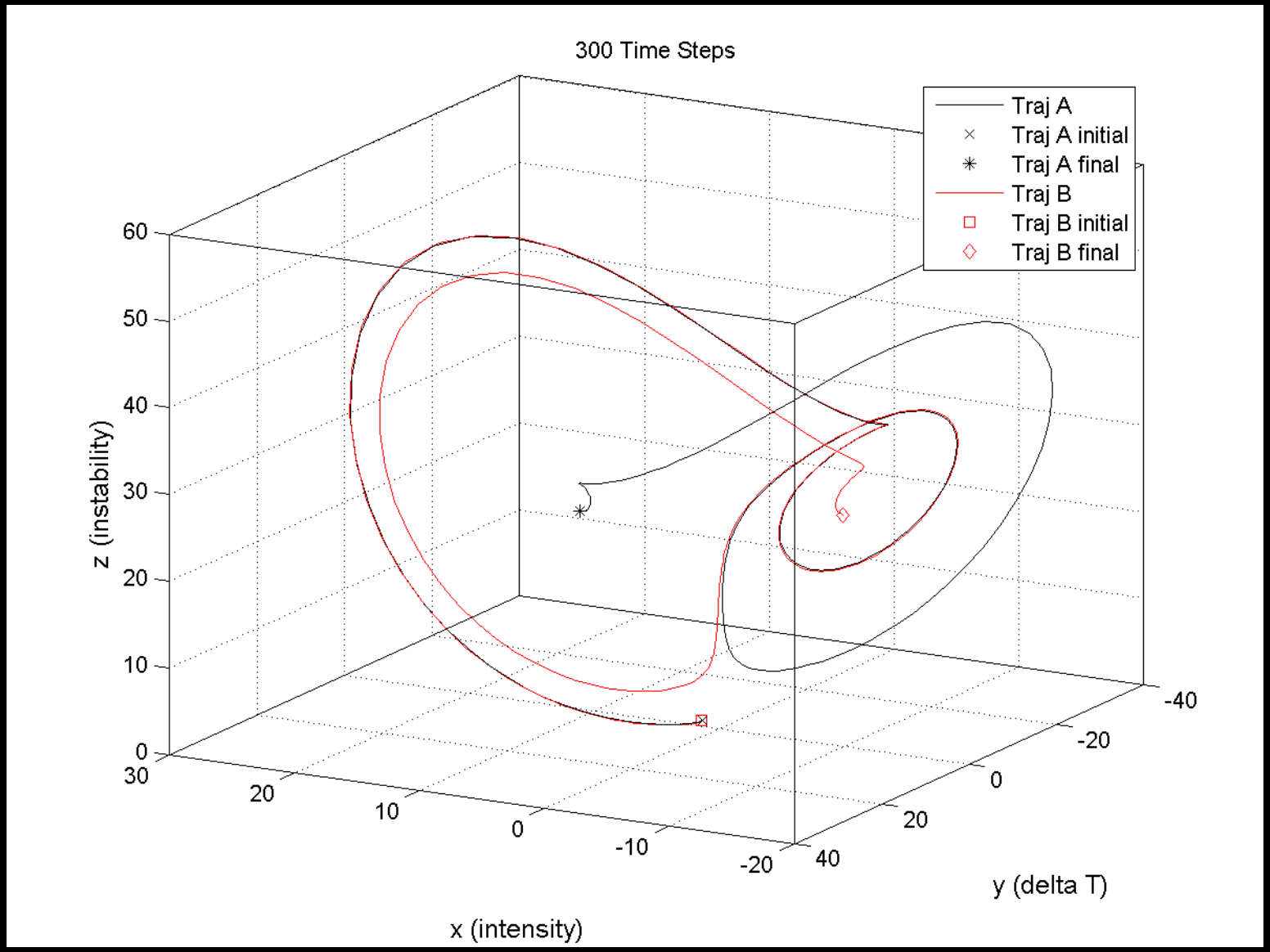


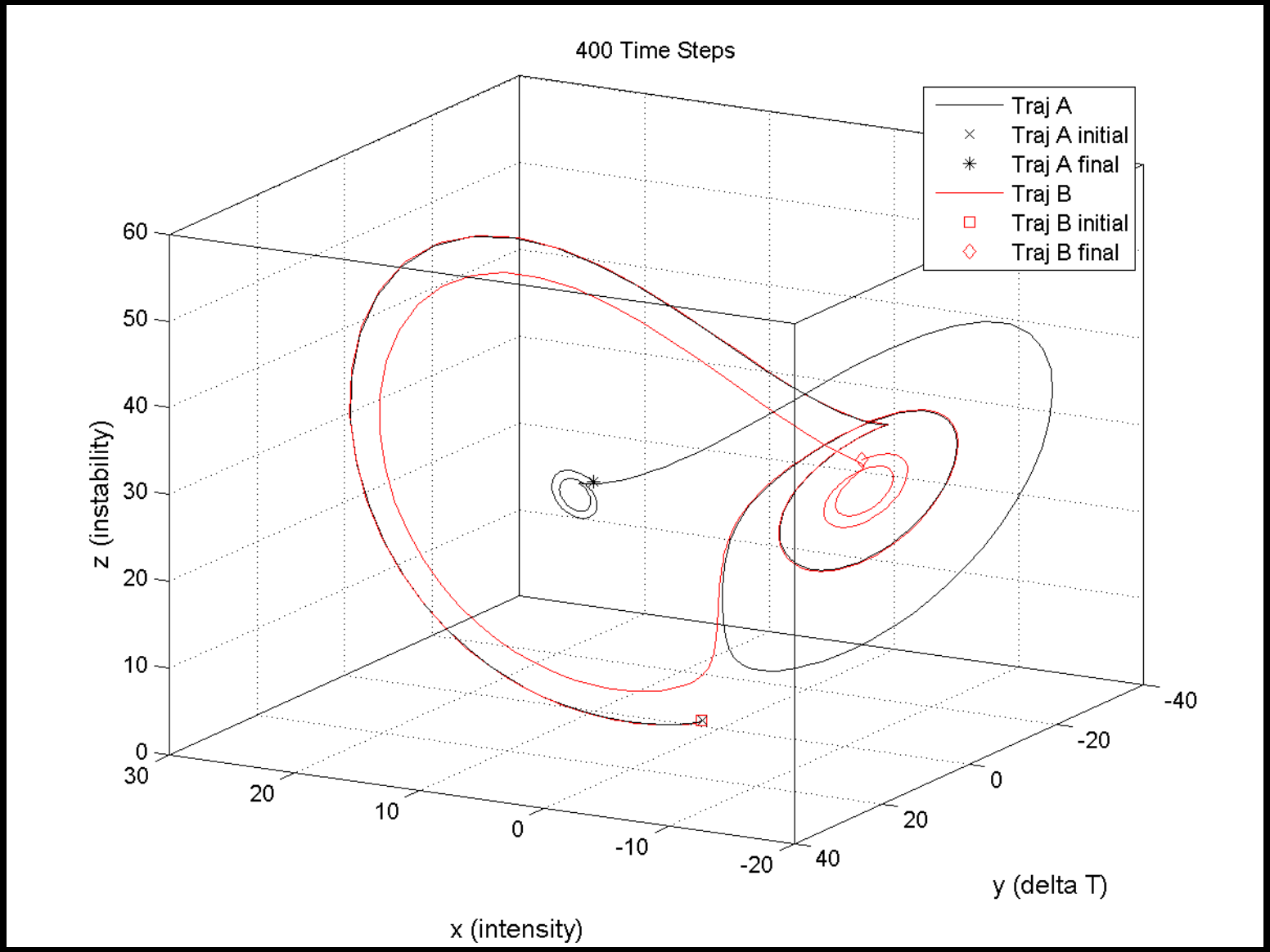
Trajectory B and **Trajectory A** have similar forecasts.

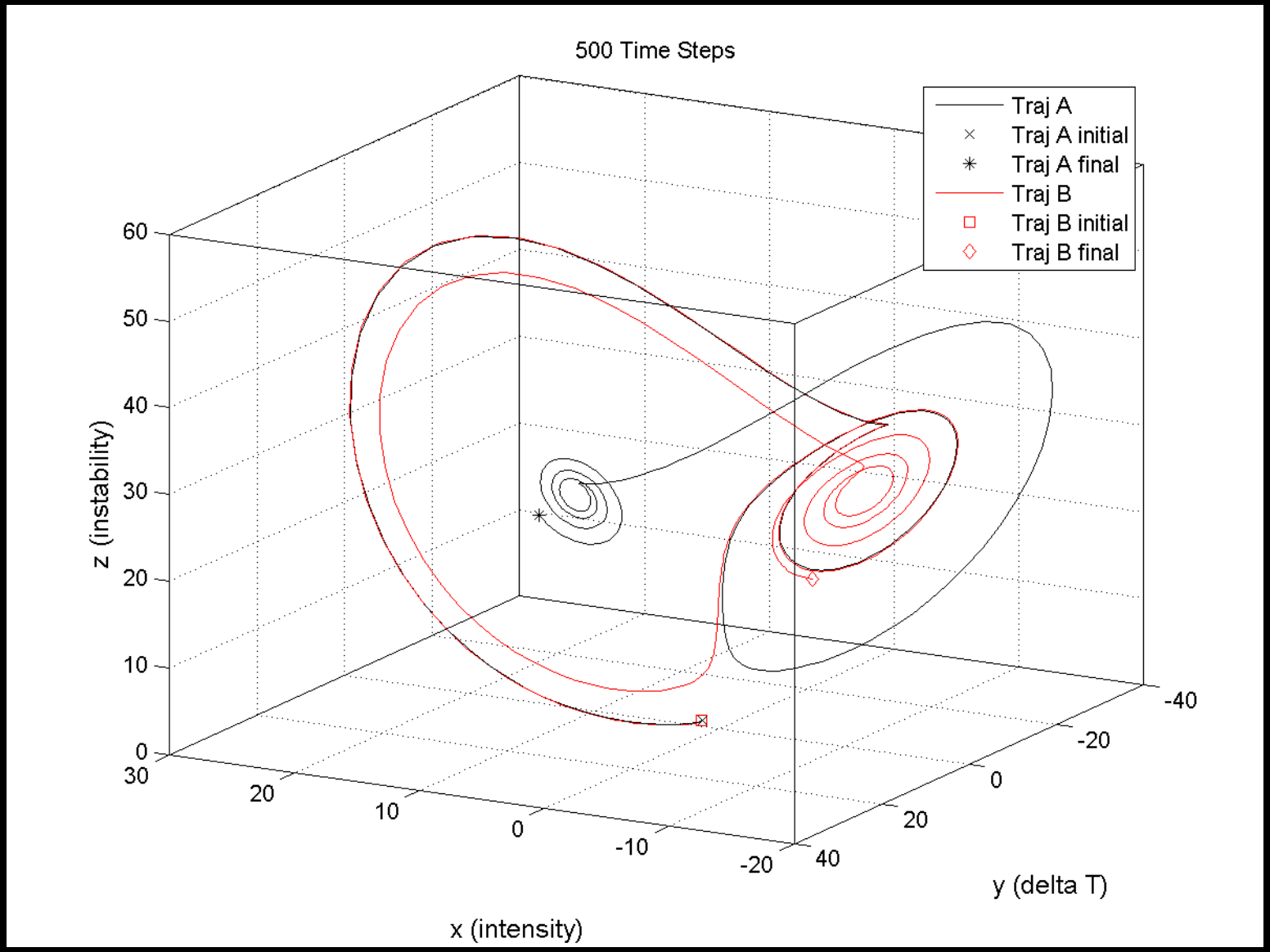


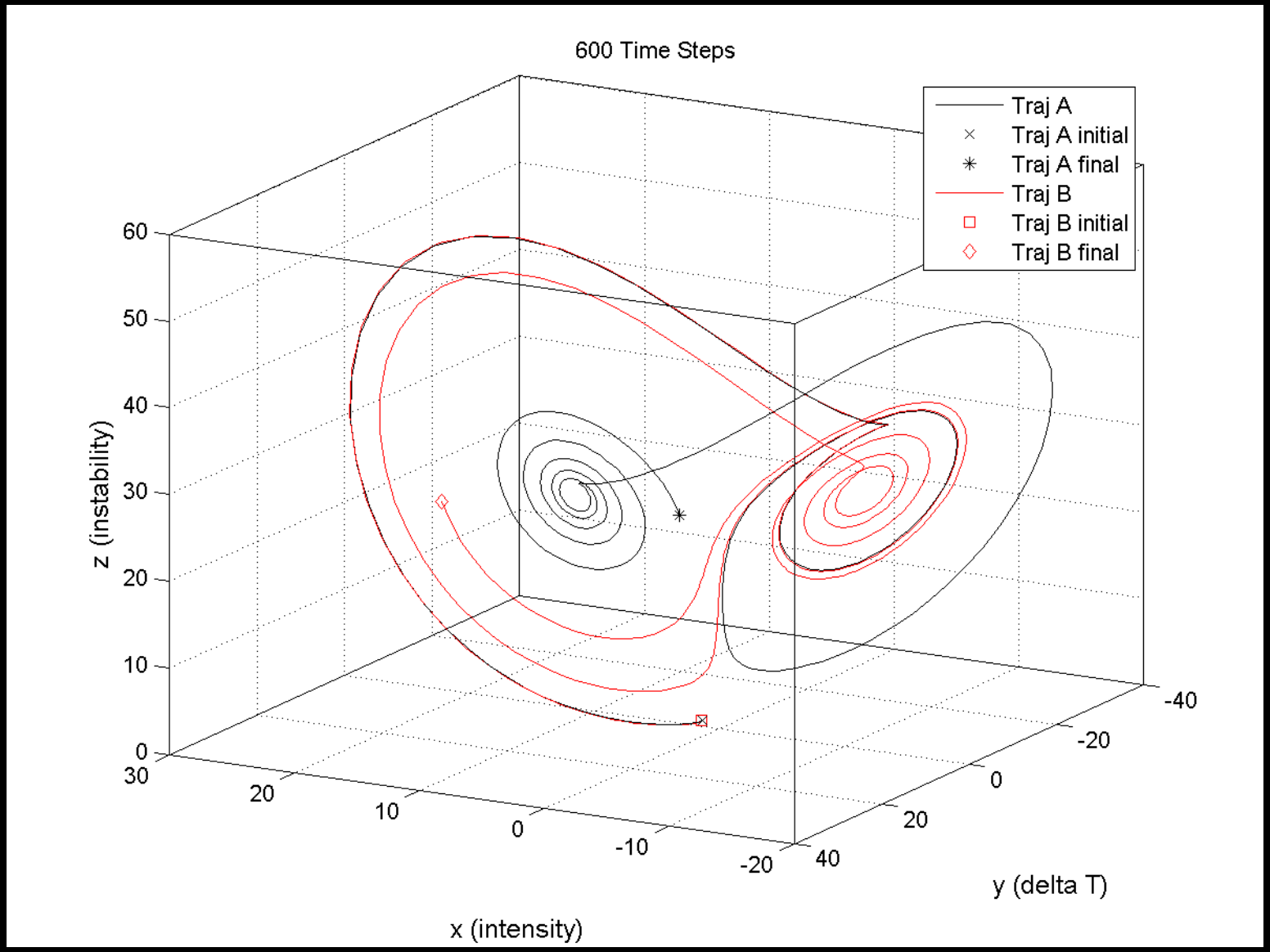
Trajectory B and **Trajectory A** forecasts are now quite different!

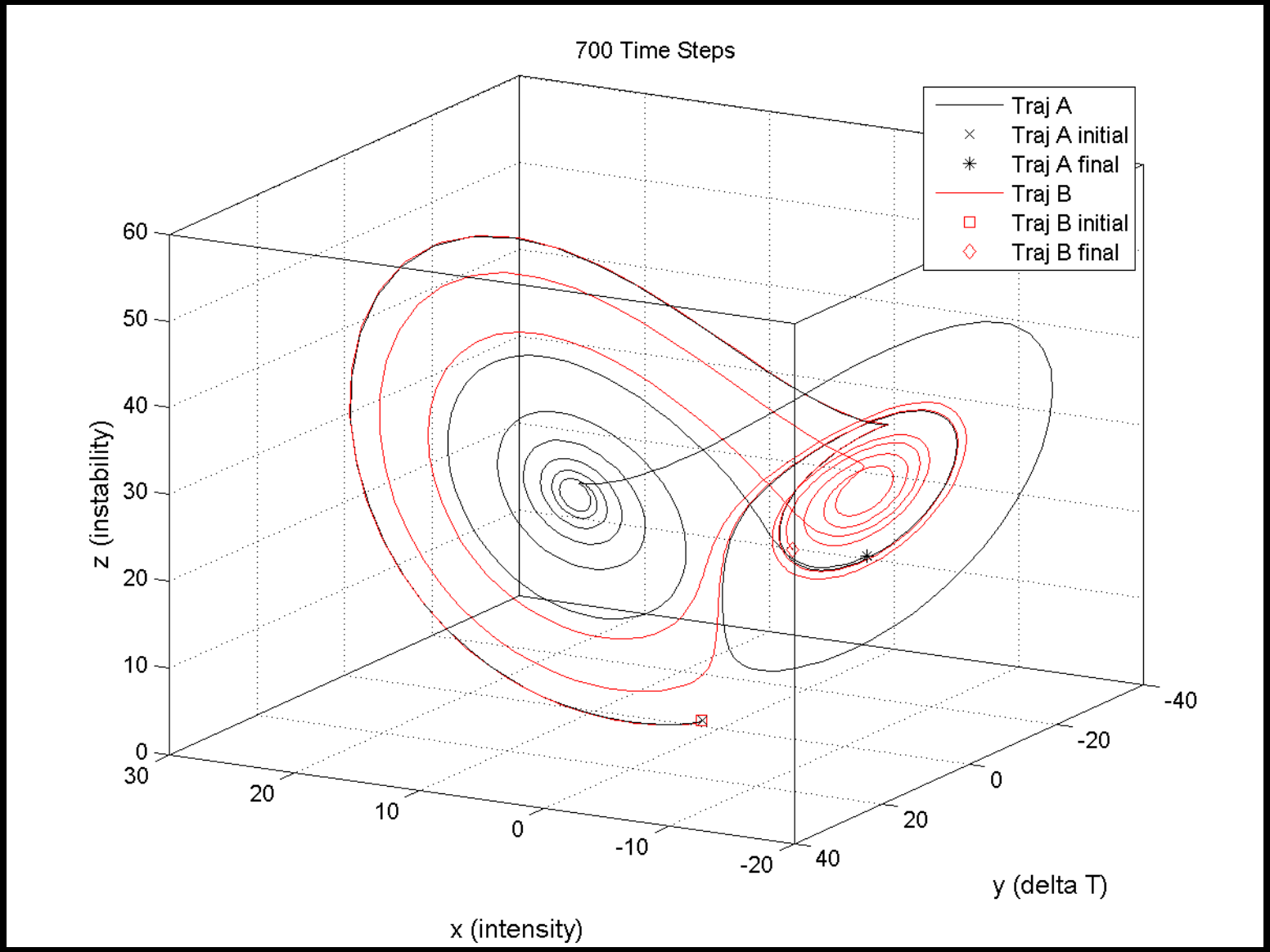

250 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)

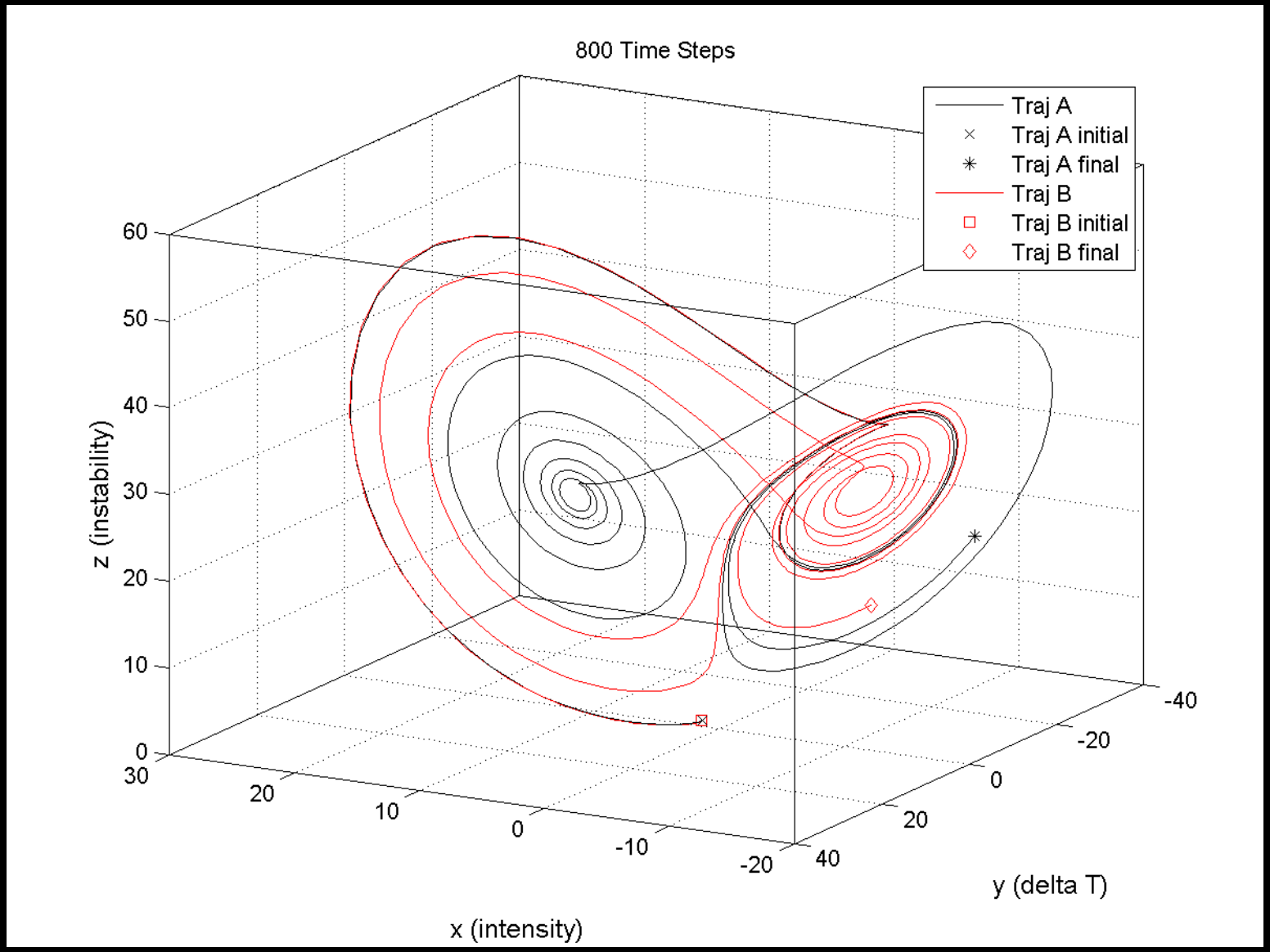

300 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)

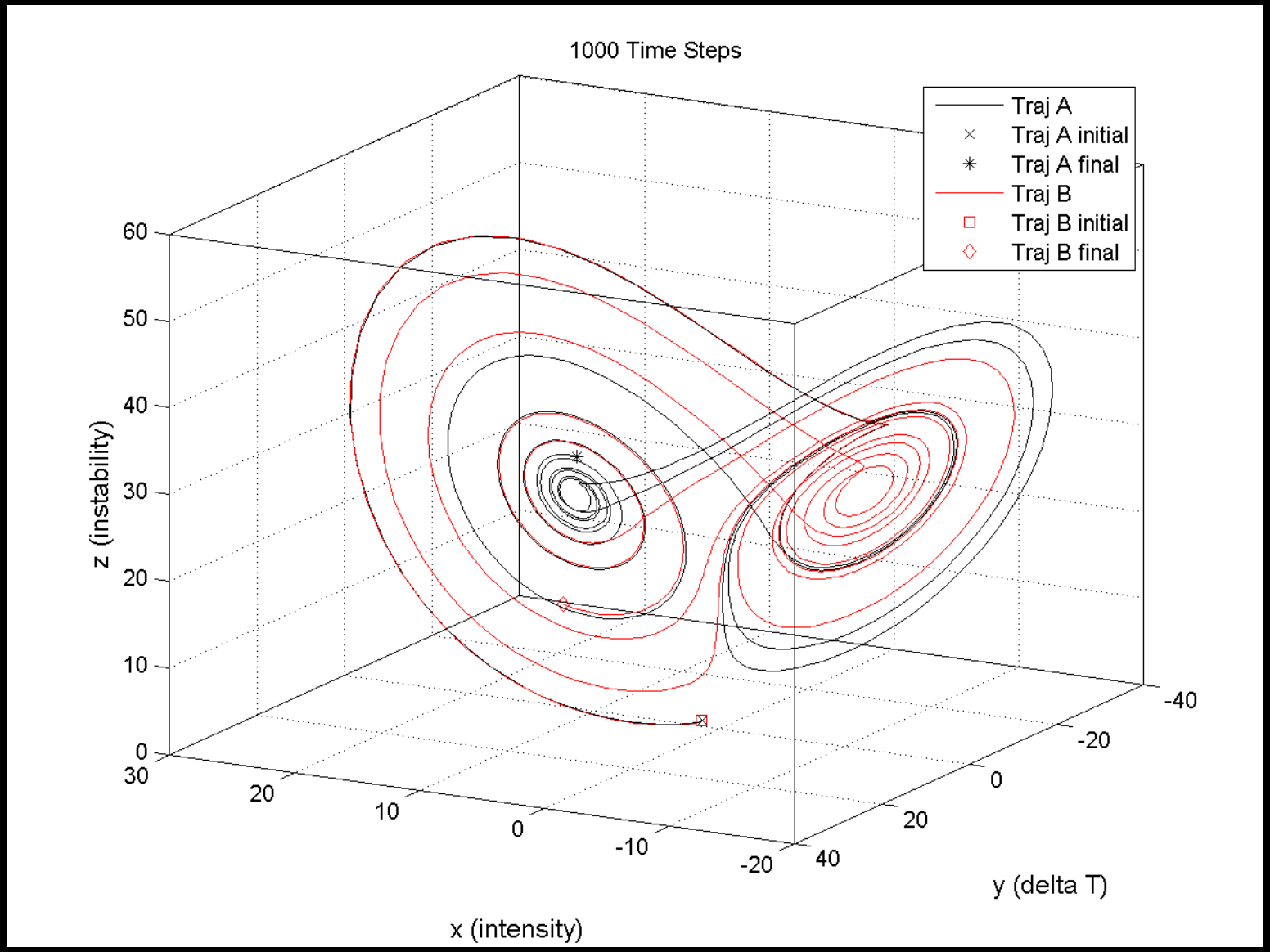

400 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)

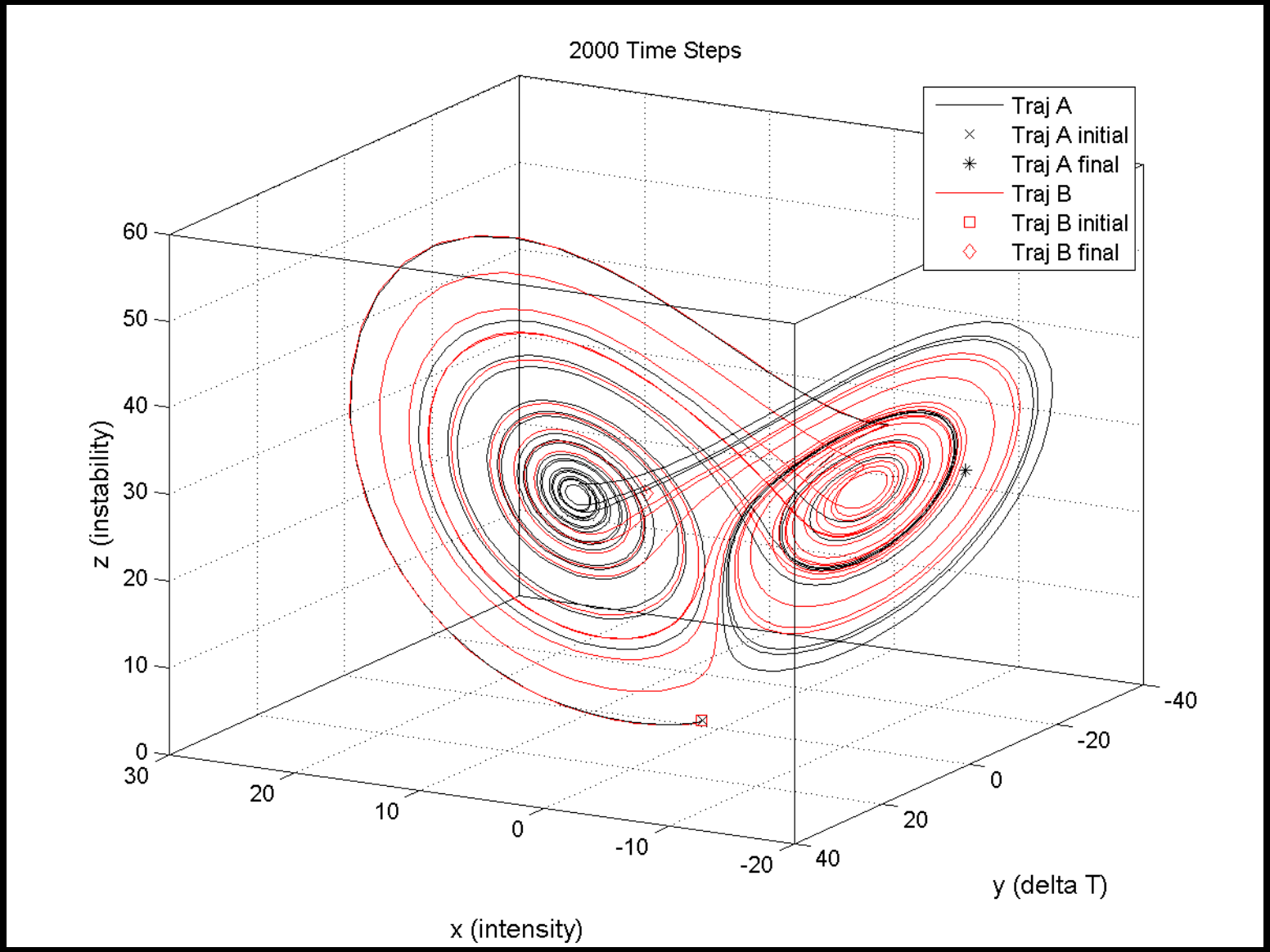

500 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)

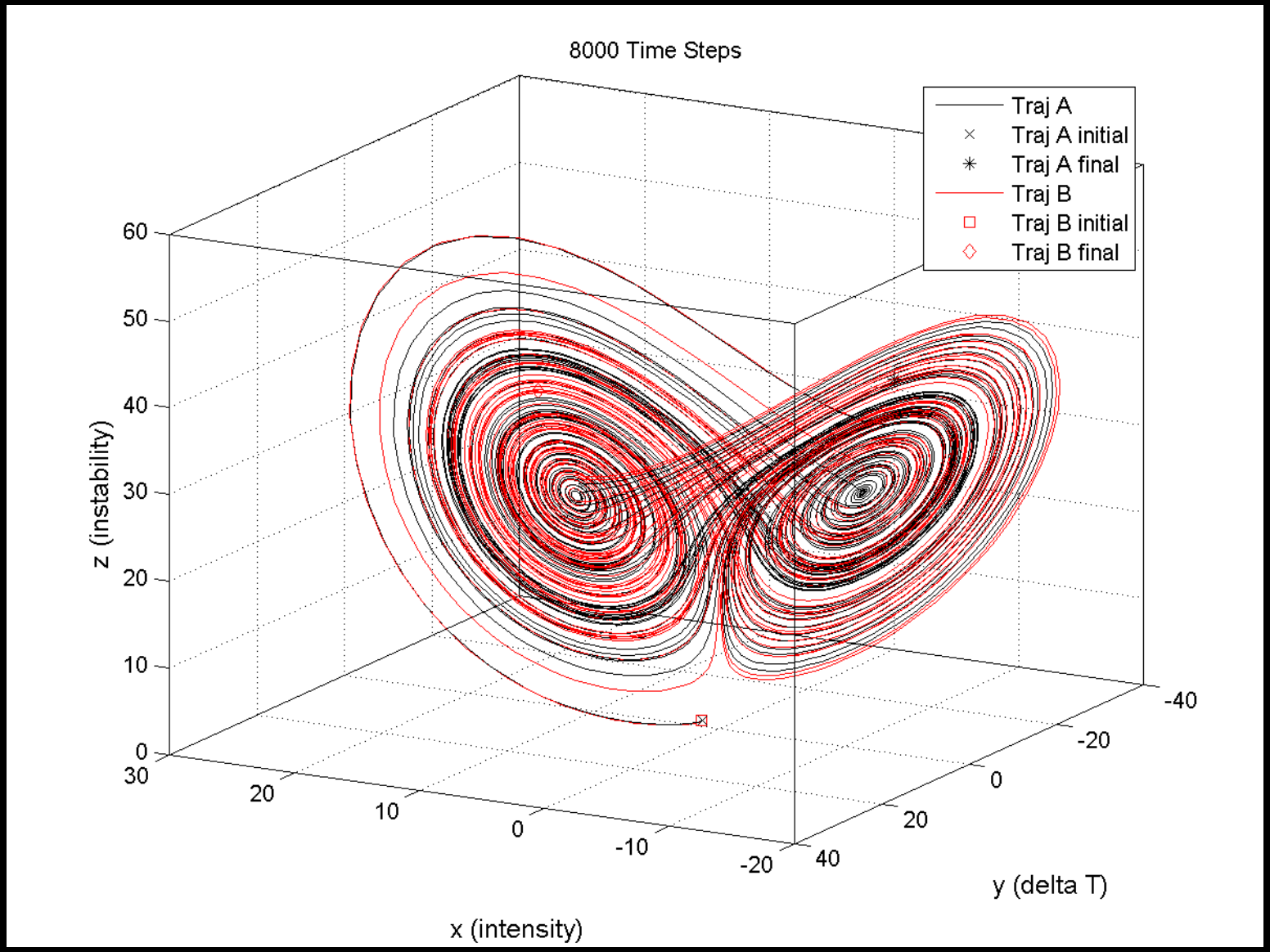

600 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)

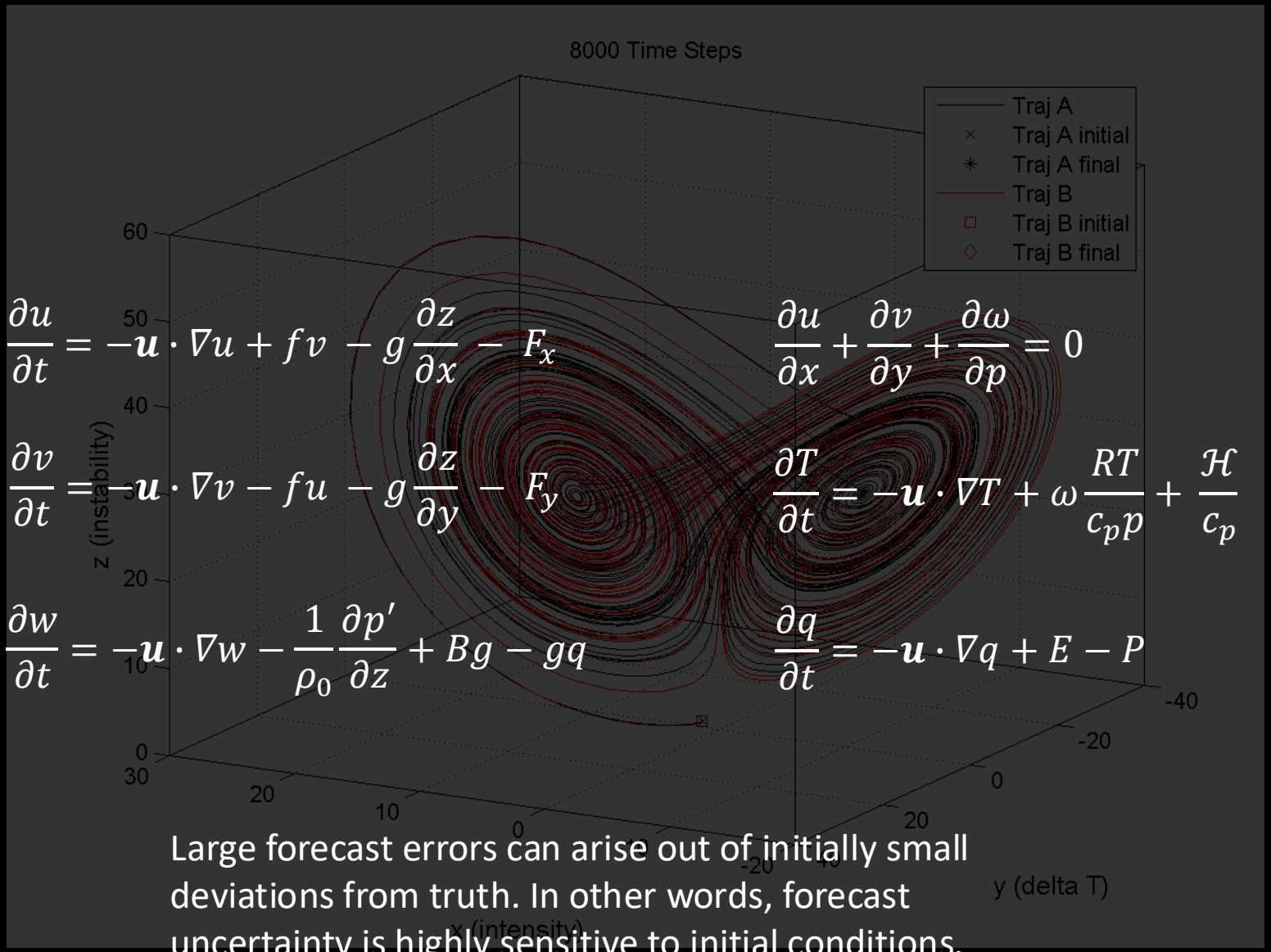

700 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)


800 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)


1000 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)


2000 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)


8000 Time Steps
Traj A
Traj A initial
Traj A final
Traj B
Traj B initial
Traj B final
z (instability)
60
50
40
30
20
10
0
30
20
10
0
-10
-20
x (intensity)
40
20
0
-20
-40
y (delta T)



$$\frac{\partial u}{\partial t} = -\boldsymbol{u} \cdot \nabla u + fv - g\frac{\partial z}{\partial x} - F_x$$

$$\frac{\partial v}{\partial t} = -\boldsymbol{u} \cdot \nabla v - fu - g\frac{\partial z}{\partial y} - F_y$$

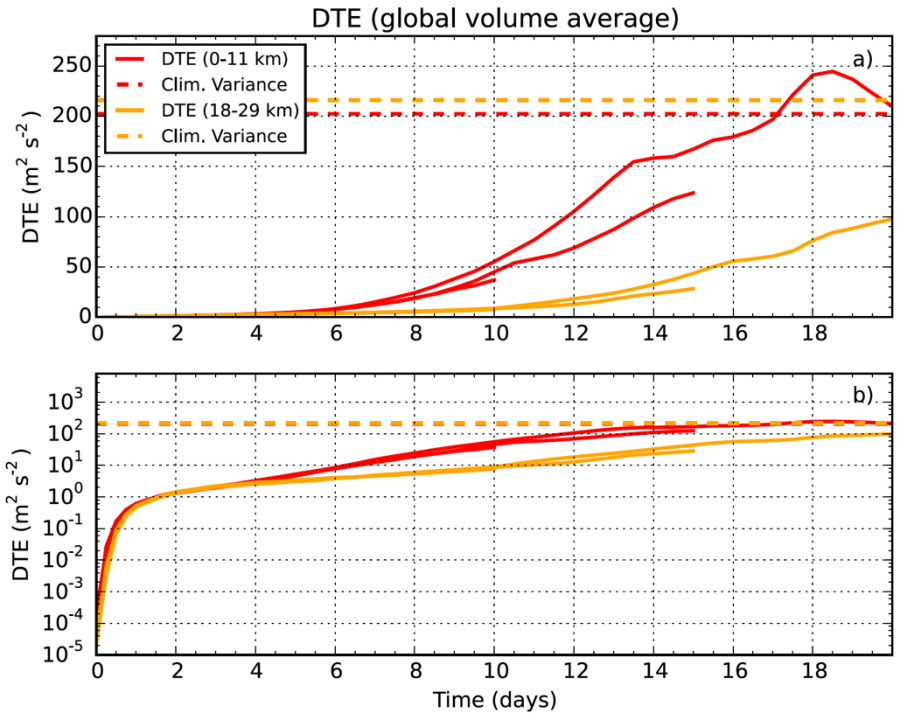
$$\frac{\partial w}{\partial t} = -\boldsymbol{u} \cdot \nabla w - \frac{1}{\rho_0}\frac{\partial p'}{\partial z} + Bg - gq$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial \omega}{\partial p} = 0$$

$$\frac{\partial T}{\partial t} = -\boldsymbol{u} \cdot \nabla T + \omega\frac{RT}{c_p p} + \frac{\mathcal{H}}{c_p}$$

$$\frac{\partial q}{\partial t} = -\boldsymbol{u} \cdot \nabla q + E - P$$

Large forecast errors can arise out of initially small deviations from truth. In other words, forecast uncertainty is highly sensitive to initial conditions.



DTE = Difference Total Energy

DKE = Difference Kinetic Energy of 10-m wind
Z500 = 500 mb geopotential height

Modern limits of predictability for NWP systems are between 2 and 3 weeks.

Judt (2018)

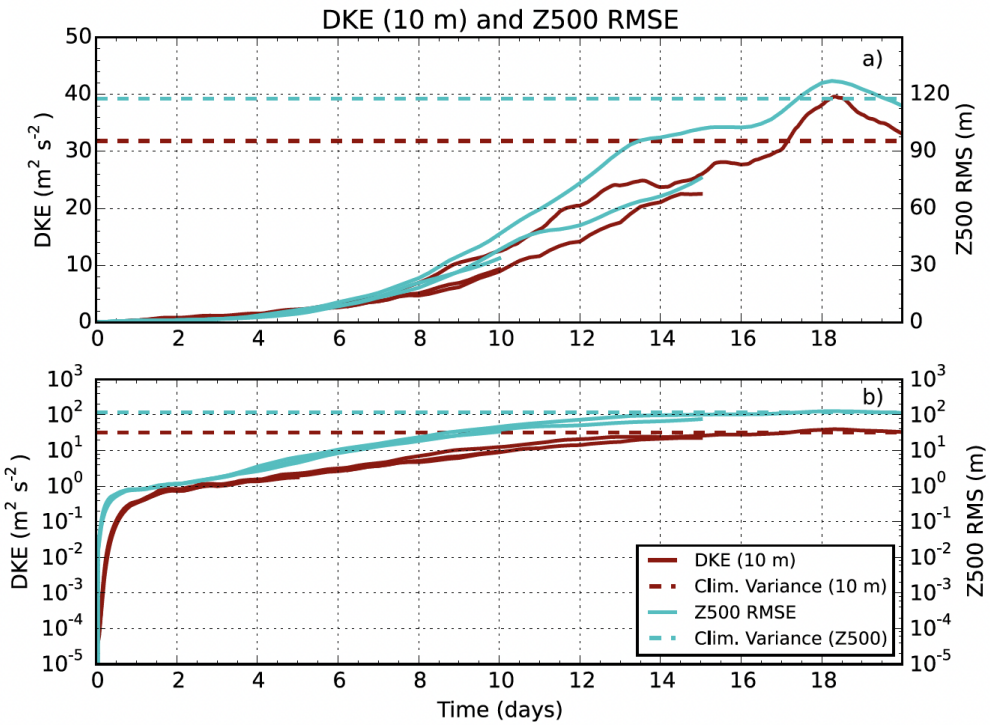


FIG. 10. As in Fig. 8, but for DKE_{10m} and $Z500_{RMSE}$.